



Calculus AB Practice Exam and Notes

Updated

Rationale C for question 30 has been corrected.

Important Note

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Effective Fall 2016

A A

CALCULUS AB
SECTION I, Part A

Time—60 minutes

Number of questions—30

NO CALCULATOR IS ALLOWED FOR THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and place the letter of your choice in the corresponding box on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

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- (A) 2.95 (B) 3.95 (C) 4.05 (D) 4.1

- (A) Zero (B) One (C) Two (D) Three

A A

7. What is the total area of the regions between the curves $y = 6x^2 - 18x$ and $y = -6x$ from $x = 1$ to $x = 3$?
- (A) 4 (B) 12 (C) 16 (D) 20

8. The function g is defined by $g(x) = x^2 + bx$, where b is a constant. If the line tangent to the graph of g at $x = -1$ is parallel to the line that contains the points $(0, -2)$ and $(3, 4)$, what is the value of b ?
- (A) -1 (B) 2 (C) $\frac{5}{2}$ (D) 4

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$$f(x) = \begin{cases} \frac{|x|}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

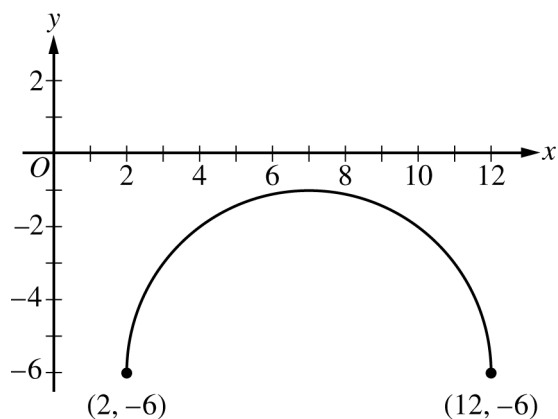
9. The function f is defined above. The value of $\int_{-5}^3 f(x) \, dx$ is

- (A) -2 (B) 2 (C) 8 (D) nonexistent

10. Let g be a continuous function. Using the substitution $u = 2x - 1$, the integral $\int_2^3 g(2x - 1) \, dx$ is equal to which of the following?

- (A) $\int_2^3 g(u) \, du$ (B) $\frac{1}{2} \int_2^3 g(u) \, du$ (C) $\int_3^5 g(u) \, du$ (D) $\frac{1}{2} \int_3^5 g(u) \, du$

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Graph of f

11. The graph of $y = f(x)$ consists of a semicircle with endpoints at $(2, -6)$ and $(12, -6)$, as shown in the figure above. What is the value of $\int_2^{12} f(x) \, dx$?

(A) $-\frac{25\pi}{2}$ (B) $\frac{25\pi}{2}$ (C) $-60 + \frac{25\pi}{2}$ (D) $60 - \frac{25\pi}{2}$

GO ON TO THE NEXT PAGE.

14. The graph of the function f , shown above, consists of three line segments. If the function g is an antiderivative of f such that $g(2) = 5$, for how many values of c , where $0 \leq c \leq 6$, does $g(c) = 3$?
- (A) Zero (B) One (C) Two (D) Three

$$f(x) = \begin{cases} 6 + cx & \text{for } x < 1 \\ 9 + 2 \ln x & \text{for } x \geq 1 \end{cases}$$

15. Let f be the function defined above, where c is a constant. If f is continuous at $x = 1$, what is the value of c ?
- (A) 2 (B) 3 (C) 5 (D) 9

GO ON TO THE NEXT PAGE.

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- (D) $y'' - y' + 5 = 0$

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{4}{7}$ (D) 2

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-2	-6	9	-10	16
1	5	-3	3	-2
3	0	7	8	3

21. The table above gives values of f , f' , g , and g' for selected values of x . If $h(x) = f(g(x))$, what is the value of $h'(1)$?

- (A) -19 (B) -14 (C) 7 (D) 9

22. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = \frac{x+1}{y}$ with the initial condition $f(0) = -2$. Which of the following is an expression for $f(x)$?

- (A) $-2 - \sqrt{x^2 + 2x}$
 (B) $-2 + \sqrt{x^2 + 2x}$
 (C) $-\sqrt{x^2 + 2x + 4}$
 (D) $\sqrt{x^2 + 2x + 4}$

GO ON TO THE NEXT PAGE.

- (A) $\pi \int_0^4 (x - (x - 2)^2) dx$
- (B) $\pi \int_0^4 (\sqrt{x} - (x - 2))^2 dx$
- (C) $\pi \int_0^2 x dx + \pi \int_2^4 (x - (x - 2)^2) dx$
- (D) $\pi \int_0^2 x dx + \pi \int_2^4 (\sqrt{x} - (x - 2))^2 dx$

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26. $\int_1^2 \frac{x^2 - x - 5}{x + 2} dx =$

-
- A diagram of a hemispherical bowl of radius 6 m. The bowl is partially filled with water to a depth h . The water surface is a smaller circle inside the bowl's rim. The radius of the bowl is labeled as 6 m, and the depth of the water is labeled as h .

- AP Calculus AB Practice Exam 27

- (A) $\frac{19}{2}$
(B) $2 \ln 2 + 9$
(C) $2 \ln 4 + 8$
(D) $2 \ln 16 + 2$

- (A) $-\frac{1}{16}$ (B) $-\frac{4}{81}$ (C) $\frac{1}{4}$ (D) 4

30. Which of the following limits is equal to $\int_2^5 x^2 dx$?

(A) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2 + \frac{k}{n}\right)^2 \frac{1}{n}$

(B) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2 + \frac{k}{n}\right)^2 \frac{3}{n}$

(C) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2 + \frac{3k}{n}\right)^2 \frac{1}{n}$

(D) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2 + \frac{3k}{n}\right)^2 \frac{3}{n}$

END OF PART A

**IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON PART A ONLY.**

DO NOT GO ON TO PART B UNTIL YOU ARE TOLD TO DO SO.

B**B****B****B****B****B****B****B****B**

CALCULUS AB
SECTION I, Part B

Time—45 minutes
Number of questions—15

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and place the letter of your choice in the corresponding box on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

In this exam:

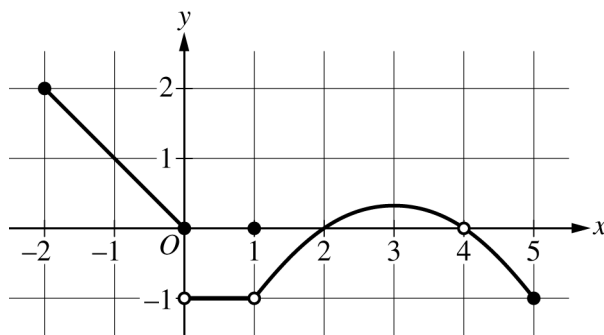
- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

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B**B****B****B****B****B****B****B****B**

76. To help restore a beach, sand is being added to the beach at a rate of $s(t) = 65 + 24 \sin(0.3t)$ tons per hour, where t is measured in hours since 5:00 A.M. How many tons of sand are added to the beach over the 3-hour period from 7:00 A.M. to 10:00 A.M.?

(A) 255.368 (B) 225.271 (C) 85.123 (D) 10.388



Graph of f

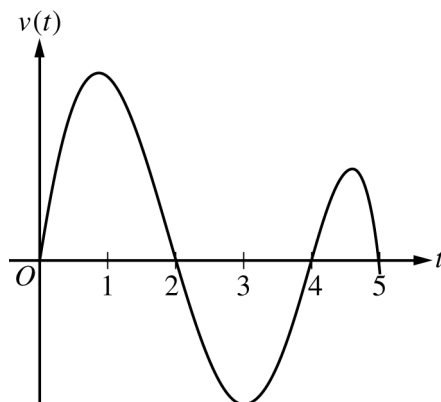
77. The graph of the function f is shown above. For what values of a does $\lim_{x \rightarrow a} f(x) = 0$?
- (A) 2 only
(B) 2 and 4
(C) 0 and 2 only
(D) 0, 1, and 2

GO ON TO THE NEXT PAGE.

B**B****B****B****B****B****B****B****B**

78. The second derivative of a function f is given by $f''(x) = \sin(3x) - \cos(x^2)$. How many points of inflection does the graph of f have on the interval $0 < x < 3$?

- (A) One (B) Three (C) Four (D) Five



79. Over the time interval $0 \leq t \leq 5$, a particle moves along the x -axis. The graph of the particle's velocity, v , is shown above. Over the time interval $0 \leq t \leq 5$, the particle's displacement is 3 and the particle travels a total distance of 13. What is the value of $\int_2^4 v(t) \, dt$?

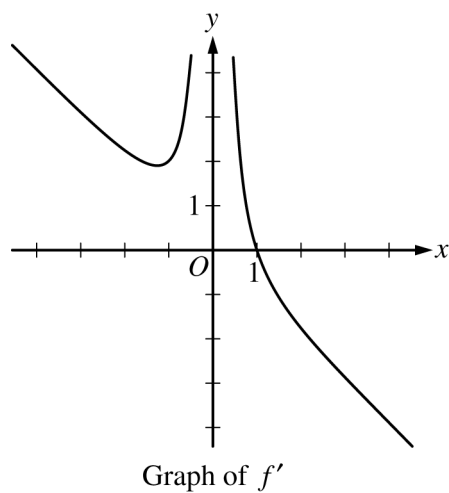
- (A) -10 (B) -5 (C) 5 (D) 10

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B**B****B****B****B****B****B****B****B**

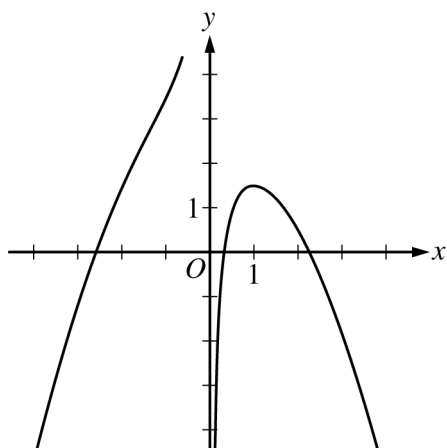
80. The temperature in a room at midnight is 20 degrees Celsius. Over the next 24 hours, the temperature changes at a rate modeled by the differentiable function H , where $H(t)$ is measured in degrees Celsius per hour and time t is measured in hours since midnight. Which of the following is the best interpretation of $\int_0^6 H(t) dt$?
- (A) The temperature of the room, in degrees Celsius, at 6:00 A.M.
 - (B) The average temperature of the room, in degrees Celsius, between midnight and 6:00 A.M.
 - (C) The change in the temperature of the room, in degrees Celsius, between midnight and 6:00 A.M.
 - (D) The rate at which the temperature in the room is changing, in degrees Celsius per hour, at 6:00 A.M.

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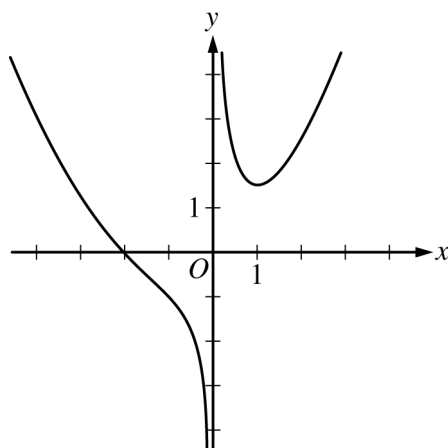
B**B****B****B****B****B****B****B****B**

81. The graph of f' , the derivative of the function f , is shown above. Which of the following could be the graph of f ?

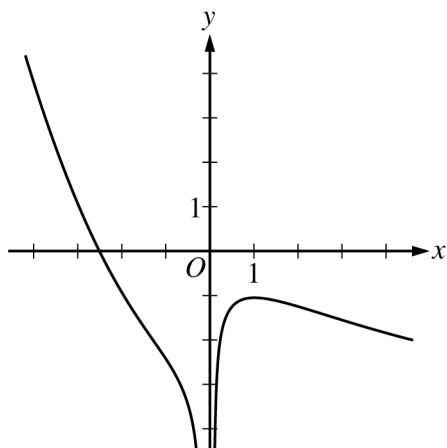
(A)



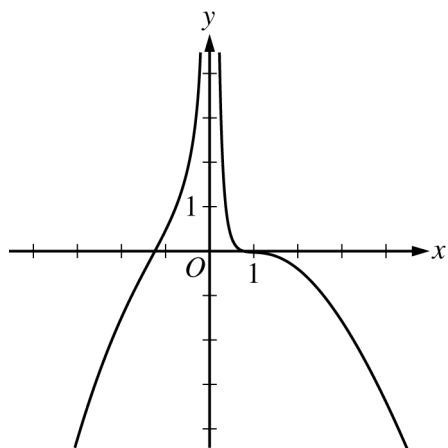
(B)



(C)



(D)

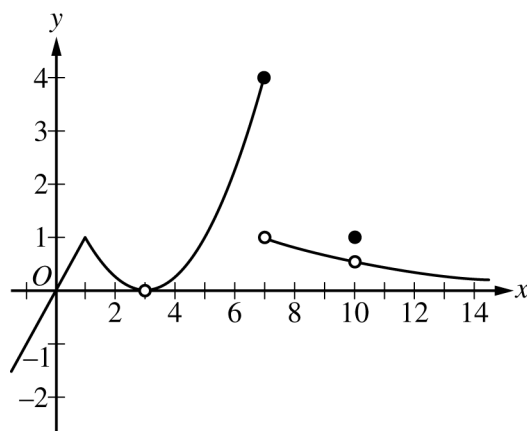


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B**B****B****B****B****B****B****B****B**

82. Let f be the function with derivative given by $f'(x) = \sin(x^2 - 3)$. At what values of x in the interval $-3 < x < 3$ does f have a relative maximum?

- (A) -1.732 and 2.478 only
- (B) -2.478 and 1.732 only
- (C) -2.138 , 0 , and 2.138
- (D) -2.478 , -1.732 , 1.732 , and 2.478



Graph of f

83. The graph of the function f is shown above. At what value of x does f have a jump discontinuity?

- (A) 1
- (B) 3
- (C) 7
- (D) 10

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B**B****B****B****B****B****B****B****B**

84. Let f be a differentiable function such that $f(1) = \pi$ and $f'(x) = \sqrt{x^3 + 6}$. What is the value of $f(5)$?
- (A) 11.941 (B) 14.587 (C) 24.672 (D) 27.814

-
85. People are entering a building at a rate modeled by $f(t)$ people per hour and exiting the building at a rate modeled by $g(t)$ people per hour, where t is measured in hours. The functions f and g are nonnegative and differentiable for all times t . Which of the following inequalities indicates that the rate of change of the number of people in the building is increasing at time t ?
- (A) $f(t) > 0$
(B) $f'(t) > 0$
(C) $f(t) - g(t) > 0$
(D) $f'(t) - g'(t) > 0$

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B**B****B****B****B****B****B****B****B**

86. The velocity of a particle moving along the x -axis is given by $v(t) = \sqrt{t} - \cos(e^t)$ for $t \geq 0$. Which of the following statements describes the motion of the particle at $t = 1$?

- (A) The particle is moving to the left with positive acceleration.
- (B) The particle is moving to the right with positive acceleration.
- (C) The particle is moving to the left with negative acceleration.
- (D) The particle is moving to the right with negative acceleration.

87. A tire that is leaking air has an initial air pressure of 30 pounds per square inch (psi). The function $t = f(p)$ models the amount of time t , in hours, it takes for the air pressure of the tire to reach p psi. What are the units for $f'(p)$?

- (A) hours (B) psi (C) psi per hour (D) hours per psi

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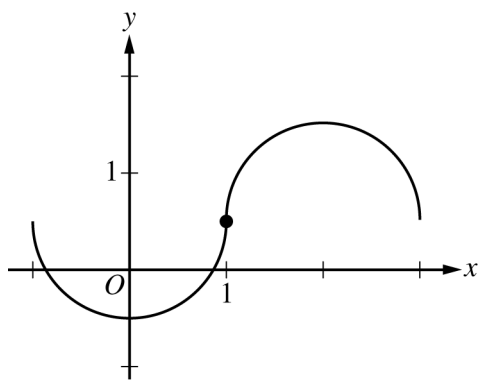
B**B****B****B****B****B****B****B****B**

88. The first derivative of the function f is defined by $f'(x) = \frac{x + 2e^{-x}}{x^2 + 0.7}$. On what intervals is f increasing?
- (A) $-1.384 < x < -0.264$ only
- (B) $x < -0.633$ and $x > 0.319$ only
- (C) $-\infty < x < \infty$
- (D) There are no intervals on which f is increasing.

x	0	4	6	8	13
$f(x)$	3	4.5	3	2.5	4.4

89. The table above shows selected values of a continuous function f . For $0 \leq x \leq 13$, what is the fewest possible number of times $f(x) = 4$?
- (A) One (B) Two (C) Three (D) Four

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B**B****B****B****B****B****B****B****B**

Graph of h'

90. The function h is defined on the closed interval $[-1, 3]$. The graph of h' , the derivative of h , is shown above. The graph consists of two semicircles with a common endpoint at $x = 1$. Which of the following statements about h must be true?

I. $h(-1) = h(3)$

II. h is continuous at $x = 1$.

III. The graph of h has a vertical asymptote at $x = 1$.

- (A) None (B) II only (C) I and II only (D) I and III only

END OF SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON PART B ONLY.**

DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

CALCULUS AB
SECTION II, Part A
Time—30 minutes
Number of problems—2

A GRAPHING CALCULATOR IS REQUIRED FOR THESE PROBLEMS.

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1**1****1****1****1****1****1****1****1****1**

t (minutes)	0	2	5	7	10
$h(t)$ (inches)	3.5	10.0	15.5	18.5	20.0

1. The depth of water in tank A , in inches, is modeled by a differentiable and increasing function h for $0 \leq t \leq 10$, where t is measured in minutes. Values of $h(t)$ for selected values of t are given in the table above.

- (a) Use the data in the table to find an approximation for $h'(6)$. Show the computations that lead to your answer. Indicate units of measure.

-
- (b) Approximate the value of $\int_0^{10} h(t) \, dt$ using a right Riemann sum with the four subintervals indicated by the data in the table. Is this approximation greater than or less than $\int_0^{10} h(t) \, dt$? Give a reason for your answer.

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1**1****1****1****1****1****1****1****1****1**

- (c) The depth of water in tank B , in inches, is modeled by the function $g(t) = 3.2 + 17.5\sqrt{\sin(0.16t)}$ for $0 \leq t \leq 10$, where t is measured in minutes. Find the average depth of the water in tank B over the interval $0 \leq t \leq 10$. Is this value greater than or less than the average depth of the water in tank A over the interval $0 \leq t \leq 10$? Give a reason for your answer.

-
- (d) According to the model given in part (c), is the depth of the water in tank B increasing or decreasing at time $t = 6$? Give a reason for your answer.

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2. Particle Q moves along the x -axis so that its velocity at any time t is given by $v_Q(t) = 1 - 3 \cos\left(\frac{t^2}{5}\right)$, and its acceleration at any time t is given by $a_Q(t) = \frac{6t}{5} \sin\left(\frac{t^2}{5}\right)$. The particle is at position $x = 2$ at time $t = 0$.
- (a) In the interval $0 < t < 5$, when is the velocity of particle Q increasing? Give a reason for your answer.

-
- (b) Find the position of particle Q at time $t = 3$.

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2**2****2****2****2****2****2****2****2****2**

- (c) A second particle, R , moves along the x -axis so that its position at any time t is given by a differentiable function $x_R(t)$, where $x_R(1) = 4$ and $x_R(3) = 8$. Explain why there must be a time t , for $1 < t < 3$, at which the velocity of particle R is 2.

-
- (d) At time $t = 3$, the velocity of particle R described in part (c) is -2 . Are particles Q and R moving toward each other or away from each other at time $t = 3$? Explain your reasoning.

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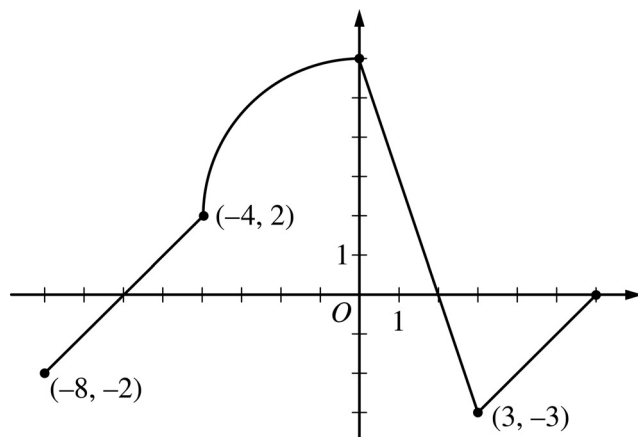
CALCULUS AB
SECTION II, Part B
Time—60 minutes
Number of problems—4

NO CALCULATOR IS ALLOWED FOR THESE PROBLEMS.

DO NOT BEGIN PART B UNTIL YOU ARE TOLD TO DO SO.

GO ON TO THE NEXT PAGE.

NO CALCULATOR ALLOWED

Graph of g

3. A continuous function g is defined on the closed interval $-8 \leq x \leq 6$. The graph of g , shown above, consists of three line segments and a quarter of a circle centered at the point $(0, 2)$. Let f be the function given by

$$f(x) = \int_{-8}^x g(t) \, dt.$$

- (a) Find all values of x in the interval $-8 < x < 6$ at which f has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

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NO CALCULATOR ALLOWED

(b) Find $f(0)$.

(c) Find $\lim_{x \rightarrow -4} \frac{f(x)}{x^2 + 4x}$.

(d) Let h be the function defined by $h(x) = \frac{g(x)}{x^2 + 1}$. Find $h'(1)$.

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NO CALCULATOR ALLOWED

4. Consider the differential equation $\frac{dy}{dx} = (y - 2)(x^2 + 1)$.

(a) Find $y = g(x)$, the particular solution to the given differential equation with initial condition $g(0) = 5$.

(b) For the particular solution $y = g(x)$ found in part (a), find $\lim_{x \rightarrow -\infty} g(x)$.

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NO CALCULATOR ALLOWED

- (c) Let $y = f(x)$ be the particular solution to the given differential equation with initial condition $f(1) = 3$.

Find the value of $\frac{d^2y}{dx^2}$ at the point $(1, 3)$. Is the graph of $y = f(x)$ concave up or concave down at the point $(1, 3)$? Give a reason for your answer.

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5**5****5****5****5****5****5****5****5****5****NO CALCULATOR ALLOWED**

5. The function f is defined by

$$f(x) = \begin{cases} 3x^2 + 2x & \text{for } x \leq 0 \\ e^{2x} + 2 & \text{for } x > 0. \end{cases}$$

- (a) Is f continuous at $x = 0$? Justify your answer.

-
- (b) Find $f'(-2)$ and $f'(3)$.

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NO CALCULATOR ALLOWED

(c) Explain why $f'(0)$ does not exist.

(d) Let g be the function given by $g(x) = \int_{-1}^x f(t) dt$. Find $g(1)$.

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NO CALCULATOR ALLOWED

6. A hive contains 35 hundred bees at time $t = 0$. During the time interval $0 \leq t \leq 4$ hours, bees enter the hive at a rate modeled by $E(t) = 16t - 3t^2$, where $E(t)$ is measured in hundreds of bees per hour. During the same time interval, bees leave the hive at a rate modeled by $L(t) = -2t + 15$, where $L(t)$ is measured in hundreds of bees per hour.
- (a) How many bees leave the hive during the time interval $0 \leq t \leq 2$?

-
- (b) Write an expression involving one or more integrals for the total number of bees, in hundreds, in the hive at time t for $0 \leq t \leq 4$. Find the total number of bees in the hive at $t = 4$.

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6**6****6****6****6****6****6****6****6****6****NO CALCULATOR ALLOWED**

- (c) Find the minimum number of bees in the hive for $0 \leq t \leq 4$. Justify your answer.

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Answers to Multiple-Choice Questions

Part A	1 – B
	2 – D
	3 – B
	4 – B
	5 – C
	6 – A
	7 – B
	8 – D
	9 – A
	10 – D
	11 – C
	12 – D
	13 – A
	14 – D
	15 – B
	16 – D
	17 – A
	18 – D
	19 – B
	20 – C
	21 – B
	22 – C
	23 – C
	24 – C
	25 – B
	26 – A
	27 – A
	28 – C
	29 – C
	30 – D

Part B	76 – A
	77 – B
	78 – D
	79 – B
	80 – C
	81 – A
	82 – A
	83 – C
	84 – D
	85 – D
	86 – B
	87 – D
	88 – C
	89 – C
	90 – B

Section I: Multiple Choice

The multiple-choice section on each exam is designed for broad coverage of the course content. Multiple-choice questions are discrete, as opposed to appearing in question sets, and the questions do not appear in the order in which topics are addressed in the curriculum framework. There are 30 multiple-choice questions in Part A and 15 multiple-choice questions in Part B; students may use a graphing calculator only for Part B. Each part of the multiple-choice section is timed and students may not return to questions in Part A of the multiple-choice section once they have begun Part B.

Curriculum Framework Alignment and Rationales for Answers

Question 1

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.3B(b) Evaluate definite integrals.	3.3B2: If f is continuous on the interval $[a, b]$ and F is an antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. Lack of parentheses is a common mistake, resulting in sign errors, when the antiderivative consists of more than one term: $\left. \frac{x^3}{3} - x^2 \right _{-1}^3 = \frac{27}{3} - 9 - \left(-\frac{1}{3} \right) - 1 = 0 + \frac{1}{3} - 1 = -\frac{2}{3}.$ This answer might also be obtained if the student uses parentheses correctly but makes an algebraic mistake by taking $(-1)^2 = -1$ to get: $\left. \frac{x^3}{3} - x^2 \right _{-1}^3 = \left(\frac{27}{3} - 9 \right) - \left(-\frac{1}{3} - (-1) \right) = 0 + \left(\frac{1}{3} - 1 \right) = -\frac{2}{3}.$	
(B)	This option is correct. This question involves using the basic power rule for antidifferentiation and then correctly substituting the endpoints and evaluating: $\left. \frac{x^3}{3} - x^2 \right _{-1}^3 = \left(\frac{27}{3} - 9 \right) - \left(-\frac{1}{3} - 1 \right) = \frac{4}{3}.$	
(C)	This option is incorrect. If the process of antidifferentiation and differentiation are confused in evaluating the definite integral, the result is: $2x - 2 \Big _{-1}^3 = 8.$	

(D)	This option is incorrect. A possible error in applying the power rule for antiderivatives is to not divide by the new exponent. This error would give: $x^3 - 2x^2 \Big _{-1}^3 = 12.$
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Question 2

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1C Calculate derivatives.	2.1C3: Sums, differences, products, and quotients of functions can be differentiated using derivative rules.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. Subtraction was used in the product rule instead of addition. $f'(x) = 2e^{2x}(x^3 + 1) - e^{2x}(3x^2)$ $f'(2) = 18e^4 - 12e^4 = 6e^4$	
(B)	This option is incorrect. The correct product rule was used but the chain rule is ignored in the derivative of e^{2x}, thereby a factor of 2 was lost. $f'(x) = e^{2x}(x^3 + 1) + e^{2x}(3x^2)$ $f'(2) = 9e^4 + 12e^4 = 21e^4$	
(C)	This option is incorrect. The chain rule was correctly used but the product rule is incorrectly stated as being $\frac{d}{dx}(f(x)g(x)) = f'(x)g'(x)$. $f'(x) = 2e^{2x}(3x^2)$ $f'(2) = 2e^4(3 \cdot 2^2) = 24e^4$	
(D)	This option is correct. Use a combination of the product rule and the chain rule to compute the derivative, then evaluate that derivative at $x = 2$. $f'(x) = 2e^{2x}(x^3 + 1) + e^{2x}(3x^2)$ $f'(2) = 18e^4 + 12e^4 = 30e^4$	

Question 3

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3B Solve problems involving the slope of a tangent line.	2.3B2: The tangent line is the graph of a locally linear approximation of the function near the point of tangency.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
(A)	This option is incorrect. An incorrect equation was used, namely $y = f(a) + f'(a)x$, in constructing the equation of a line given a slope and a point on the line. $f(2.1) \approx f(2) - \frac{1}{2}(2.1) = 4 - 1.05 = 2.95$	
(B)	This option is correct. An equation of the tangent line at $x = a$ is $y = f(a) + f'(a)(x - a)$. In this question $a = 2$ and $f'(a) = -\frac{1}{2}$. The value of y when $x = 2.1$ would be an approximation to $f(2.1)$. $f(2.1) \approx f(2) - \frac{1}{2}(2.1 - 2) = 4 - \frac{1}{2}(0.1) = 4 - 0.05 = 3.95$	
(C)	This option is incorrect. An error was made in constructing the equation of a line given a slope and a point on the line, using $y = f(a) - f'(a)(x - a)$ for the equation of the tangent line. $f(2.1) \approx f(2) + \frac{1}{2}(2.1 - 2) = 4 + \frac{1}{2}(0.1) = 4.05$ The same answer could also be obtained with the correct tangent line equation by making a multiplication error or sign error in substituting the derivative value or performing the calculation.	
(D)	This option is incorrect. This is a fundamental conceptual error in not recognizing the derivative as the slope of the tangent line. $f(2.1) \approx f(2) + (2.1 - 2) = 4 + 0.1 = 4.1$	

Question 4

Learning Objective		Essential Knowledge	Mathematical Practices for AP Calculus
2.2A Use derivatives to analyze properties of a function.		2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is incorrect. The error comes from believing that g is always increasing because the derivative is the sum of two terms and is thus always positive.		
(B)	This option is correct. First find the critical values where the derivative is 0 or undefined. A relative extrema occurs at a critical value if the derivative changes sign there. $g'(x) = 4x^3 + 12x^2 = x^2(4x + 12)$ $x^2(4x + 12) = 0 \text{ at } x = 0 \text{ or } x = -3$ There are two zeros, but $g'(x)$ changes sign for $x = -3$ only. The sign of $g'(x)$ at $x = 0$ does not change because of the x^2 factor.		
(C)	This option is incorrect. The number of critical values has been confused with the number of relative extrema. There are two zeros for $g'(x) = 0$, but the derivative only changes sign at one of them.		
(D)	This option is incorrect. The error is the result of assuming that a quartic polynomial always has 3 relative extrema or that a cubic polynomial always has 3 zeros.		

Question 5

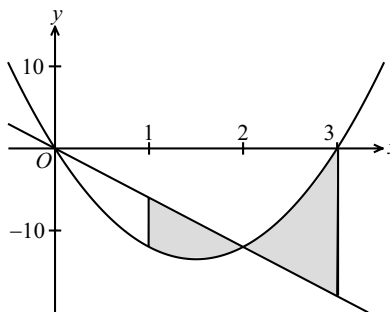
Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.4B Apply definite integrals to problems involving the average value of a function.	3.4B1: The average value of a function f over an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$.	MPAC 1: Reasoning with definitions and theorems MPAC 3: Implementing algebraic/computational processes
(A)	This option is incorrect. The average velocity has been confused with the average rate of change of velocity over the interval: $\frac{v(3) - v(1)}{3 - 1} = \frac{-7 - 1}{2} = -4.$	
(B)	This option is incorrect. The average velocity is not the arithmetic mean of the velocity values at the endpoints: $\frac{v(3) + v(1)}{2} = \frac{-7 + 1}{2} = -3.$	
(C)	This option is correct. The average value of $v(t)$ over the interval $[1, 3]$ is computed with a definite integral: $\begin{aligned} \frac{1}{3-1} \int_1^3 v(t) dt &= \frac{1}{3-1} \int_1^3 (2 - t^2) dt = \frac{1}{2} \left(2t - \frac{1}{3}t^3 \right) \Big _1^3 \\ &= \frac{1}{2} \left((6 - 9) - \left(2 - \frac{1}{3} \right) \right) = -\frac{7}{3}. \end{aligned}$	
(D)	This option is incorrect. Speed and velocity have been confused and it was assumed that the average velocity must be positive: $\left \frac{1}{3-1} \int_1^3 (2 - t^2) dt \right = \frac{7}{3}.$	

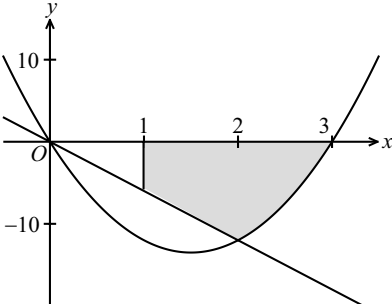
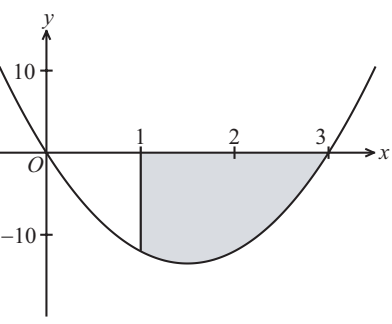
Question 6

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.2B Approximate a definite integral.	3.2B2: Definite integrals can be approximated using a left Riemann sum, a right Riemann sum, a midpoint Riemann sum, or a trapezoidal sum; approximations can be computed using either uniform or nonuniform partitions.	MPAC 1: Reasoning with definitions and theorems MPAC 3: Implementing algebraic/computational processes
(A)	This option is correct. The trapezoidal sum is the average of the left and right Riemann sums. The three intervals are of length 2, 5, and 2, respectively. Taking the average of the left and right endpoint values on each interval and multiplying by the length of the interval gives the following trapezoidal sum: $\frac{2}{2}(15 + 9) + \frac{5}{2}(9 + 5) + \frac{2}{2}(5 + 4) = 68.$	
(B)	This option is incorrect. Using the trapezoidal rule is not appropriate because the intervals are of different lengths. Ignoring the values of t and believing that $\Delta t = \frac{9 - 0}{3} = 3$ as if the intervals were of equal length in the table would yield the following computation using the trapezoidal rule: $\frac{3}{2}[15 + 2(9) + 2(5) + 4] = 70.5.$	
(C)	This option is incorrect. A trapezoidal sum has been confused with a left Riemann sum: $2 \cdot 15 + 5 \cdot 9 + 2 \cdot 5 = 85.$	
(D)	This option is incorrect. The error is due to taking just the sum of the left and right Riemann sums rather than the average, thus not dividing each width by 2: $2(15 + 9) + 5(9 + 5) + 2(5 + 4) = 136.$	

Question 7

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.4D Apply definite integrals to problems involving area, volume, (BC) and length of a curve.	3.4D1: Areas of certain regions in the plane can be calculated with definite integrals. (BC) Areas bounded by polar curves can be calculated with definite integrals.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
3.3B(b) Evaluate definite integrals.	3.3B2: If f is continuous on the interval $[a, b]$ and F is an antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$.	
(A)	This option is incorrect. If a sketch is not made or if it is not checked first whether the two graphs intersect on the interval $1 \leq x \leq 3$, it is likely that the calculation for area will be done with just one integral: $\int_1^3 (f(x) - g(x)) dx = \int_1^3 (6x^2 - 12x) dx$ $= (2x^3 - 6x^2) \Big _1^3 = (54 - 54) - (2 - 6).$ If the integration had been done as $\int_1^3 (g(x) - f(x)) dx$, the answer would be -4. But since area must be positive, the absolute value would give the same answer as the first integral.	
(B)	This option is correct. It is helpful to sketch the graphs of the two functions and to check if the two graphs cross inside the given interval. The first graph is a concave up parabola with zeros at $x = 0$ and $x = 3$, and the second graph is a line passing through the origin. The sketch indicates that there are two regions between the curves on the interval $1 \leq x \leq 3$. The line intersects the parabola where $6x^2 - 18x = -6x$, or $0 = 6x^2 - 12x = 6x(x - 2)$; therefore the top and bottom curves of the regions switch roles at $x = 2$. Different integrals should be used to find the total area of the regions, one for the interval $1 \leq x \leq 2$ and one for the interval $2 \leq x \leq 3$. Let $f(x) = 6x^2 - 18x$ and $g(x) = -6x$. The total area is: $\int_1^3 f(x) - g(x) dx = \int_1^2 (g(x) - f(x)) dx + \int_2^3 (f(x) - g(x)) dx$ $= \int_1^2 (12x - 6x^2) dx + \int_2^3 (6x^2 - 12x) dx = (6x^2 - 2x^3) \Big _1^2 + (2x^3 - 6x^2) \Big _2^3$ $= ((24 - 16) - (6 - 2)) + ((54 - 54) - (16 - 24)) = 4 + 8 = 12.$	

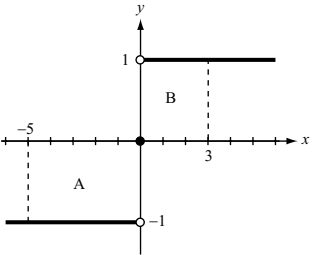


(C)	This option is incorrect. The error comes from finding the area bounded by the curves and the x-axis between $x = 1$ and $x = 3$. Since the region is below the axis, the area is the opposite of the integral value: $\left(-\int_1^2 g(x) \, dx\right) + \left(-\int_2^3 f(x) \, dx\right).$	 The graph shows a coordinate plane with a parabola $f(x)$ opening upwards and a line $g(x)$ with a negative slope. They intersect at the origin O and at $x = 3$. The region between the curves from $x = 1$ to $x = 3$ is shaded. The parabola is above the line in this interval, and both are below the x-axis.
(D)	This option is incorrect. The error comes from finding the area bounded only by the parabola and x-axis between $x = 1$ and $x = 3$. Because the region is below the axis, the area is the opposite of the integral value: $-\int_1^3 f(x) \, dx.$	 The graph shows the same parabola $f(x)$ as in the previous graph, intersecting the x-axis at $x = 0$ and $x = 3$. The region between the parabola and the x-axis from $x = 1$ to $x = 3$ is shaded. The parabola is below the x-axis in this interval.

Question 8

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3B Solve problems involving the slope of a tangent line.	2.3B1: The derivative at a point is the slope of the line tangent to a graph at that point on the graph.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is incorrect. The slope of the tangent line has been incorrectly set equal to $g(-1)$ instead of $g'(-1)$: $g(-1) = (-1)^2 + b(-1) = 2$ $\Rightarrow b = -1.$	
(B)	This option is incorrect. Only the slope of the line was calculated through the two given points, but that was not connected to the calculation to find the value of b .	
(C)	This option is incorrect. The slope of the line through $(0, -2)$ and $(3, 4)$ was mistakenly found using the wrong difference quotient, $\frac{\Delta x}{\Delta y} = \frac{1}{2}.$ $g'(x) = 2x + b$ $g'(-1) = 2(-1) + b = \frac{1}{2} \Rightarrow b = \frac{5}{2}$	
(D)	This option is correct. Parallel lines have the same slope. The slope of the tangent line at $x = -1$ is therefore equal to the slope of the line through $(0, -2)$ and $(3, 4)$, which is $\frac{4 - (-2)}{3 - 0} = 2$. Therefore find b so that $g'(-1) = 2.$ $g'(x) = 2x + b$ $g'(-1) = 2(-1) + b = 2 \Rightarrow b = 4$	

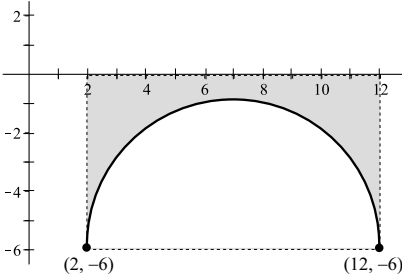
Question 9

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.2C Calculate a definite integral using areas and properties of definite integrals.	3.2C3: The definition of the definite integral may be extended to functions with removable or jump discontinuities.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is correct. This function has a jump discontinuity at $x = 0$. $f(x) = -1$ for $x < 0$ and $f(x) = 1$ for $x > 0$. The graph of f is shown to the right along with the regions A and B between the graph of f and the x-axis on the intervals $[-5, 0]$ and $[0, 3]$, respectively.  Computing the definite integral in terms of areas and taking into account which region is below the axis and which is above, we get: $\int_{-5}^3 f(x) \, dx = -\text{area}(A) + \text{area}(B) = -5 + 3 = -2.$ The definite integral over the interval $[-5, 3]$ can also be written as the sum of the definite integrals over $[-5, 0]$ and $[0, 3]$, giving: $\int_{-5}^3 f(x) \, dx = \int_{-5}^0 -1 \, dx + \int_0^3 1 \, dx = -5 + 3 = -2.$	
(B)	This option is incorrect. Sign errors were made in the definition of f so that the graph of f is reflected across the x-axis. This then yields: $\int_{-5}^3 f(x) \, dx = \text{area}(A) - \text{area}(B) = 5 - 3 = 2$ or $\int_{-5}^3 f(x) \, dx = \int_{-5}^0 1 \, dx + \int_0^3 -1 \, dx = 5 - 3 = 2.$	
(C)	This option is incorrect. Rather than taking into account the signed areas, this is the total area: $\int_{-5}^3 f(x) \, dx = \text{area}(A) + \text{area}(B) = 5 + 3 = 8$ or $\int_{-5}^3 f(x) \, dx = \left \int_{-5}^0 -1 \, dx \right + \left \int_0^3 1 \, dx \right = 5 + 3 = 8.$	
(D)	This option is incorrect. The source of the error may be not recognizing that the definition of the definite integral can be extended to functions with removable or jump discontinuities.	

Question 10

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.3B(b) Evaluate definite integrals.	3.3B5: Techniques for finding antiderivatives include algebraic manipulation such as long division and completing the square, substitution of variables, (BC) integration by parts, and nonrepeating linear partial fractions.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. The technique of substitution of variables was not applied correctly because only the substitution $u = 2x - 1$ was made. There is no substitution for dx or for the limits of integration.	
(B)	This option is incorrect. The technique of substitution of variables was not applied correctly because only the substitutions for $2x - 1$ and dx were made, but not for the limits of integration.	
(C)	This option is incorrect. The technique of substitution of variables was not applied correctly because only the substitutions for $2x - 1$ and the limits of integration were made, but not for dx .	
(D)	This option is correct. Starting with the substitution $u = 2x - 1$, $u = 2x - 1 \Rightarrow du = 2dx \Rightarrow dx = \frac{1}{2} du.$ Also change the limits of integration for x to limits of integration for u: $x = 2 \Rightarrow u = 2 \cdot 2 - 1 = 3$ $x = 3 \Rightarrow u = 2 \cdot 3 - 1 = 5.$ Substituting for $2x - 1$, for dx, and for the limits of integration gives: $\int_2^3 g(2x - 1) dx = \int_3^5 g(u) \cdot \frac{1}{2} du = \frac{1}{2} \int_3^5 g(u) du.$	

Question 11

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.2C Calculate a definite integral using areas and properties of definite integrals.	3.2C1: In some cases, a definite integral can be evaluated by using geometry and the connection between the definite integral and area.	MPAC 3: Implementing algebraic/computational processes MPAC 4: Connecting multiple representations
(A)	This option is incorrect. Only the area of the semicircle was considered. The opposite of that area was then taken, since the region is below the axis.	
(B)	This option is incorrect. Only the area of the semicircle was considered, ignoring both the area of the rectangle and the fact that the region is below the axis.	
(C)	This option is correct. The definite integral can be evaluated by using geometry and the connection of the definite integral with areas. In particular, the value of the integral here is the opposite of the area of the region between the semicircle and the x-axis, i.e. the area of the rectangle with vertices at $(2, 0)$, $(12, 0)$, $(2, -6)$, and $(12, -6)$ minus the area of the semicircle. The area of the rectangle is $6 \cdot 10 = 60$. The area of the semicircle of radius 5 is $\frac{1}{2}\pi(5)^2 = \frac{25\pi}{2}$. The definite integral is therefore equal to $-\left(60 - \frac{25\pi}{2}\right)$.	
(D)	This option is incorrect. Only the area of the rectangle minus the area of the semicircle was computed; the need to take the opposite of this value, since the region is below the axis, was not recognized.	

Question 12

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.5A Analyze differential equations to obtain general and specific solutions.	3.5A1: Antidifferentiation can be used to find specific solutions to differential equations with given initial conditions, including applications to motion along a line, exponential growth and decay, (BC) and logistic growth.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is incorrect. This is the velocity at $t = 2$, not the position: $v(t) = 3t^2 + 10 \Rightarrow v(2) = 22$. The velocity at $t = 2$ can also be found directly using the Fundamental Theorem of Calculus: $v(2) = v(0) + \int_0^2 v'(t) dt = 10 + \int_0^2 a(t) dt = 10 + \int_0^2 6t dt = 22$.	
(B)	This option is incorrect. The assumption was made that the velocity is constant and the acceleration was ignored: $p(2) = 7 + 2 \cdot v(0) = 7 + 2 \cdot 10 = 27$.	
(C)	This option is incorrect. This is the total change in position without accounting for the initial position. The calculation uses the correct velocity function but takes $p(t) = t^3 + 10t$.	
(D)	This option is correct. Velocity is the antiderivative of acceleration and position is the antiderivative of velocity. In each case, the object's velocity and position at $t = 0$ can be used to find the appropriate "+ C" after each antidifferentiation. The last step is to then evaluate the position function at $t = 2$. $a(t) = 6t$ $v(t) = 3t^2 + C_1$ $v(0) = 3(0)^2 + C_1 = 10 \Rightarrow C_1 = 10$ $v(t) = 3t^2 + 10$ $p(t) = t^3 + 10t + C_2$ $p(0) = (0)^3 + 10(0) + C_2 = 7 \Rightarrow C_2 = 7$ $p(t) = t^3 + 10t + 7$ $p(2) = 2^3 + 10(2) + 7 = 35$	

Question 13

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1D Determine higher order derivatives.	2.1D1: Differentiating f' produces the second derivative f'' , provided the derivative of f' exists; repeating this process produces higher order derivatives of f .	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is correct. This question requires repeated differentiation using knowledge of the derivatives of sine and cosine, the derivative of the natural logarithm, and the power rule for derivatives, as well as recognizing the need to use the chain rule for the first derivative. $\frac{dy}{dx} = -\sin x - \frac{2}{2x} = -\sin x - \frac{1}{x}$ $\frac{d^2y}{dx^2} = -\cos x + \frac{1}{x^2}$ $\frac{d^3y}{dx^3} = \sin x - \frac{2}{x^3}$	
(B)	This option is incorrect. A consistent sign error was made with the derivatives of the two trigonometric functions by using $\frac{d}{dx}(\sin x) = -\cos x \text{ and } \frac{d}{dx}(\cos x) = \sin x.$ $\frac{dy}{dx} = \sin x - \frac{2}{2x} = \sin x - \frac{1}{x}$ $\frac{d^2y}{dx^2} = -\cos x + \frac{1}{x^2}$ $\frac{d^3y}{dx^3} = -\sin x - \frac{2}{x^3}$	

(C)	This option is incorrect. The only error was not using the chain rule when taking the first derivative. No chain rule is needed after that: $\frac{dy}{dx} = -\sin x - \frac{1}{2x}$ $\frac{d^2y}{dx^2} = -\cos x + \frac{1}{2x^2}$ $\frac{d^3y}{dx^3} = \sin x - \frac{2}{2x^3} = \sin x - \frac{1}{x^3}.$
(D)	This option is incorrect. A chain rule error was made on the first derivative, and there are consistent sign errors with the derivatives of the trigonometric functions. $\frac{dy}{dx} = \sin x - \frac{1}{2x}$ $\frac{d^2y}{dx^2} = -\cos x + \frac{1}{2x^2}$ $\frac{d^3y}{dx^3} = -\sin x - \frac{2}{2x^3} = -\sin x - \frac{1}{x^3}$

Question 14

Learning Objective		Essential Knowledge	Mathematical Practices for AP Calculus
3.3A Analyze functions defined by an integral.		3.3A3: Graphical, numerical, analytical, and verbal representations of a function f provide information about the function g defined as $g(x) = \int_a^x f(t) dt$.	MPAC 4: Connecting multiple representations MPAC 1: Reasoning with definitions and theorems
3.2C Calculate a definite integral using areas and properties of definite integrals.		3.2C1: In some cases, a definite integral can be evaluated by using geometry and the connection between the definite integral and area.	
(A)	This option is incorrect. The Fundamental Theorem of Calculus was applied to calculate values of $g(x)$, but all the areas were added to the value $g(2)$. Therefore the conclusion was that $g(x) \geq 5$ for all values of x .		
(B)	This option is incorrect. It comes from confusing the graph of g with the graph of f and finding the number of times that $f(x) = 3$, which happens only between $x = 5$ and $x = 6$.		
(C)	This option is incorrect. The Fundamental Theorem of Calculus was applied correctly to find values of $g(x)$ for $x > 2$, thus finding the solutions $c = 4$ and $5 < c < 6$. However, the integral $\int_2^0 f(x) dx$ might have been mishandled, thus missing the solution at $c = 0$.		
(D)	This option is correct. By the Fundamental Theorem of Calculus and using area and geometry to calculate the definite integral, $g(0) = g(2) + \int_2^0 g'(x) dx = 5 - \int_0^2 f(x) dx = 5 - 2 = 3.$ So $c = 0$ is one solution to $g(x) = 3$. Since $g' = f$, the graph of f indicates that g is increasing from $x = 0$ to $x = 2$ since f is positive there. From $x = 2$ to $x = 4$ the function g decreases by the same amount that it increased from $x = 0$ to $x = 2$ by the symmetry of the regions. $g(4) = 5 + \int_2^4 f(x) dx = 5 + (-2) = 3$ This gives a second solution at $c = 4$. According to the graph of the derivative f, the function g continues to decrease until $x = 5$ and then begins increasing again. $g(5) = 5 + \int_2^5 f(x) dx = 5 + (-3) = 2$ $g(6) = 2 + \int_5^6 f(x) dx = 2 + 2 = 4$ Since g is continuous, the Intermediate Value Theorem guarantees a third solution somewhere on the interval $(5, 6)$. There are no other solutions possible. (The third solution is at $c = 5.5$.)		

Question 15

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.2A Analyze functions for intervals of continuity or points of discontinuity.	1.2A1: A function f is continuous at $x = c$ provided that $f(c)$ exists, $\lim_{x \rightarrow c} f(x)$ exists, and $\lim_{x \rightarrow c} f(x) = f(c)$.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. Rather than setting as equal the left- and right-sided limits at $x = 1$, the derivatives of the two pieces were set equal at $x = 1$. $\frac{d}{dx}(6 + cx) = \frac{d}{dx}(9 + 2 \ln x)$ $c = \frac{2}{x} \Rightarrow c = 2$	
(B)	This option is correct. Since the function f is continuous at $x = 1$, then $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$. Evaluate these two limits and set them equal, then solve for c. $\lim_{x \rightarrow 1^-} f(x) = 6 + c \cdot 1 = 6 + c$ $\lim_{x \rightarrow 1^+} f(x) = 9 + 2 \ln 1 = 9$ $6 + c = 9 \Rightarrow c = 3$	
(C)	This option is incorrect. Rather than setting as equal the left- and right-sided limits at $x = 1$, the derivatives of the two pieces were set equal at $x = 1$, but a calculus error was made with respect to the derivative of a constant. $\frac{d}{dx}(6 + cx) = \frac{d}{dx}(9 + 2 \ln x)$ $6 + c = 9 + \frac{2}{x}$ $6 + c = 9 + 2 \Rightarrow c = 5$ OR The left- and right-sided limits were set equal at $x = 1$, but an algebraic error was made in evaluating $\ln 1 = 1$: $6 + c(1) = 9 + 2(1) = 11 \Rightarrow c = 5.$	
(D)	This option is incorrect. The value of the function at $x = 1$ was confused with c so that $c = f(1) = 9 + 2 \ln 1 = 9$.	

Question 16

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3F Estimate solutions to differential equations.	2.3F1: Slope fields provide visual clues to the behavior of solutions to first order differential equations.	MPAC 4: Connecting multiple representations MPAC 2: Connecting concepts
(A)	This option is incorrect. For this differential equation, the slopes of the line segments in Quadrant II must be positive since $y > 0$ and $x^2 > 0$ in that quadrant. In this slope field that does not happen, as can be observed with the segments near the bottom left of Quadrant II. In addition, all line segments along the x-axis should have positive slopes. This is not the case here. This slope field might be chosen if the squared term is not accounted for. [This is the slope field for $\frac{dy}{dx} = x + y$.]	
(B)	This option is incorrect. For this differential equation, the slopes of the line segments in Quadrant I must be positive since $y > 0$ and $x^2 > 0$ in that quadrant. In this slope field, however, all the line segments in Quadrant I have negative slopes. In addition, all line segments along the x-axis should have positive slopes. This is not the case here. This slope field might be chosen if one considers $\frac{dy}{dx} = 0$ and thinks of the differential equation as relating to the parabola $y = -x^2$. [This is the slope field for $\frac{dy}{dx} = -1.8x$.]	
(C)	This option is incorrect. For this differential equation, the slopes of the line segments in Quadrant II must be positive since $y > 0$ and $x^2 > 0$ in that quadrant. In this slope field, however, that does not happen as can be observed with the segments near the bottom left of Quadrant II. In addition, all line segments along the x-axis should have positive slopes. This is not the case here. This option might be chosen if the x and y variables are confused and one looks for the slope field for $\frac{dy}{dx} = x + y^2$.	
(D)	This option is correct. The line segments in the slope field have slopes given by $\frac{dy}{dx} = x^2 + y$ at the point (x, y). In Quadrants I and II, all slopes must be positive or zero since $y > 0$ in those quadrants and $x^2 \geq 0$. This is the only option in which that condition is true.	

Question 17

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3E Verify solutions to differential equations.	2.3E2: Derivatives can be used to verify that a function is a solution to a given differential equation.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
(A)	This option is correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equations in this problem involve both y' and y''. The correct derivatives must be computed and the algebra correctly done to verify that the differential equation is satisfied. $y' = 3e^{3x} - 5$ $y'' = 9e^{3x}$ $y'' - 3y' - 15 = 9e^{3x} - 3(3e^{3x} - 5) - 15$ $= 9e^{3x} - 9e^{3x} + 15 - 15 = 0$	
(B)	This option is incorrect. The correct derivatives were found but this differential equation may appear to be satisfied if the subtraction in the second term is not distributed correctly across the two terms inside the parentheses: $y'' - 3y' + 15 = 9e^{3x} - 3(3e^{3x} - 5) + 15 = 9e^{3x} - 9e^{3x} - 15 + 15 = 0.$	
(C)	This option is incorrect. This differential equation will appear to be satisfied if the chain rule is not used in taking the derivative of the exponential: $y' = e^{3x} - 5$ $y'' = e^{3x} \text{ (error)}$ $y'' - y' - 5 = e^{3x} - (e^{3x} - 5) - 5 = e^{3x} - e^{3x} + 5 - 5 = 0.$	
(D)	This option is incorrect. This differential equation will appear to be satisfied if the chain rule is not used in taking the derivative of the exponential and the subtraction in the second term is not distributed correctly across the two terms inside the parentheses: $y' = e^{3x} - 5$ $y'' = e^{3x} \text{ (error)}$ $y'' - y' + 5 = e^{3x} - (e^{3x} - 5) + 5 = e^{3x} - e^{3x} - 5 + 5 = 0.$	

Question 18

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1C Calculate derivatives.	2.1C2: Specific rules can be used to calculate derivatives for classes of functions, including polynomial, rational, power, exponential, logarithmic, trigonometric, and inverse trigonometric.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. The incorrect derivative, $\frac{d}{dx}\sin^{-1}x = \cos^{-1}x$, was used but the inverse cosine function was correctly evaluated to get $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$.	
(B)	This option is incorrect. The incorrect derivative, $\frac{d}{dx}\sin^{-1}x = \cos^{-1}x$, was used and the inverse cosine function was incorrectly evaluated as $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$.	
(C)	This option is incorrect. The derivative of the inverse sine function was confused with the derivative of the inverse tangent function, and therefore $\frac{1}{1+x^2}$ was used for the derivative: $\frac{1}{1+\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{1}{1+\frac{3}{4}} = \frac{4}{7}.$	
(D)	This option is correct. The derivative of the inverse sine function $\sin^{-1}x$ is $\frac{1}{\sqrt{1-x^2}}$. Since $f(x) = \sin^{-1}x$, $f'\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{\sqrt{1-\left(\frac{\sqrt{3}}{2}\right)^2}} = \frac{1}{\sqrt{\frac{1}{4}}} = 2.$	

Question 19

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1A Identify the derivative of a function as the limit of a difference quotient.	2.1A2: The instantaneous rate of change of a function at a point can be expressed by $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ or $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$, provided that the limit exists. These are common forms of the definition of the derivative and are denoted $f'(a)$.	MPAC 1: Reasoning with definitions and theorems MPAC 5: Building notational fluency
(A)	This option is incorrect. This could have been caused by observing that the numerator is zero at $x = e$ and then ignoring the denominator.	
(B)	This option is correct. The limit of this difference quotient is of the form $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ where $f(x) = x^{20} - 3x$ and $a = e$. This is one way to express the derivative of f at $x = e$. $f'(x) = \frac{d}{dx}(x^{20} - 3x) = 20x^{19} - 3$ $\Rightarrow f'(e) = 20e^{19} - 3$	
(C)	This option is incorrect. This is just the value of the function $x^{20} - 3x$ at $x = e$.	
(D)	This option is incorrect. Observing that the denominator was zero at $x = e$ could have led to the assumption of unbounded behavior while the numerator was ignored. Or it was thought that the indeterminate form $\frac{0}{0}$ produced a nonexistent result, without recognizing the limit as involving the definition of the derivative.	

Question 20

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1C Calculate derivatives.	2.1C5: The chain rule is the basis for implicit differentiation.	MPAC 3: Implementing algebraic/ computational processes MPAC 5: Building notational fluency
2.1D Determine higher order derivatives.	2.1D1: Differentiating f' produces the second derivative f'' , provided the derivative of f' exists; repeating this process produces higher order derivatives of f .	
(A)	This option is incorrect. The chain rule may not have been used. $\frac{d^2y}{dx^2} = 3y^2 \Rightarrow \frac{d^2y}{dx^2}\bigg _{(x,y)=(1,2)} = 3(4) = 12$ The x and y values could also have been reversed when substituting into the second derivative. $\frac{d^2y}{dx^2} = 3y^2 \frac{dy}{dx} = 3y^2(y^3 + 3)$ $\frac{d^2y}{dx^2}\bigg _{(x,y)=(1,2)} = 3(1)(4) = 12$	
(B)	This option is incorrect. There was an error in the power rule when differentiating y^3 resulted in $3y$. $\frac{d^2y}{dx^2} = 3y \frac{dy}{dx} = 3y(y^3 + 3)$ $\frac{d^2y}{dx^2}\bigg _{(x,y)=(1,2)} = 3(2)(2^3 + 3) = 3(2)(11) = 66$	
(C)	This option is correct. The answer is obtained by using the chain rule when doing implicit differentiation. The second derivative is found in terms of y and $\frac{dy}{dx}$, which can then be written completely in terms of y . The condition $f(1) = 2$ means that $y = 2$ when $x = 1$. $\frac{d^2y}{dx^2} = 3y^2 \frac{dy}{dx} = 3y^2(y^3 + 3)$ $\frac{d^2y}{dx^2}\bigg _{(x,y)=(1,2)} = 3 \cdot 2^2(2^3 + 3) = 3(4)(11) = 132$	

(D)	This option is incorrect. There was an error in differentiating a constant function and not getting 0 (i.e. not differentiating the 3). $\frac{d^2y}{dx^2} = (3y^2 + 3)\frac{dy}{dx} = (3y^2 + 3)(y^3 + 3)$ $\left. \frac{d^2y}{dx^2} \right _{(x,y)=(1,2)} = (3 \cdot 4 + 3)(11) = 165$
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Question 21

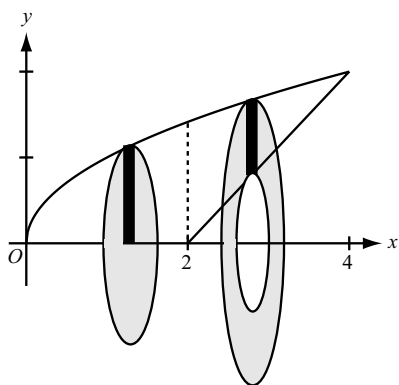
Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1C Calculate derivatives.	2.1C4: The chain rule provides a way to differentiate composite functions.	MPAC 5: Building notational fluency MPAC 4: Connecting multiple representations
(A)	This option is incorrect. The notation $f(g(x))$ was confused with the product $f(x)g(x)$ and the product rule was used to find the derivative: $h'(1) = f'(1)g(1) + g'(1)f(1)$ $h'(1) = (-3)(3) + (-2)(5) = -19.$	
(B)	This option is correct. The solution is found by using the chain rule correctly and then using values of the functions and their derivatives from the table to evaluate the derivative at $x = 1$. $h'(1) = f'(g(1))g'(1)$ $h'(1) = f'(3)g'(1)$ $h'(1) = 7(-2) = -14$	
(C)	This option is incorrect. The chain rule for composite functions was incompletely understood by thinking that $\frac{d}{dx}f(g(x)) = f'(g(x))$: $h'(1) = f'(g(1)) = 7.$	
(D)	This option is incorrect. The chain rule for composite functions was misunderstood by thinking that $\frac{d}{dx}f(g(x)) = f'(g'(x))$: $h'(1) = f'(g'(1))$ $h'(1) = f'(-2)$ $h'(1) = 9.$	

Question 22

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.5A Analyze differential equations to obtain general and specific solutions.	3.5A2: Some differential equations can be solved by separation of variables.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. The arbitrary constant must be included when doing the antidifferentiation, not after solving for y (sometimes referred to as “The <i>Late C</i>”). This is the result if the constant is introduced at the wrong time. $\frac{dy}{dx} = \frac{x+1}{y} \Rightarrow y \, dy = (x+1) \, dx$ $\int y \, dy = \int (x+1) \, dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + x \Rightarrow y^2 = x^2 + 2x$ $y = \pm\sqrt{x^2 + 2x} + C$ $-2 = 0 + C$ $y = -\sqrt{x^2 + 2x} - 2$	
(B)	This option is incorrect. The arbitrary constant was introduced after solving for y (see the steps in option (A)), and the positive square root was selected.	
(C)	This option is correct. This question involves solving a differential equation by separation of variables and using the initial condition to determine the appropriate value for the arbitrary constant. The solution leads to y^2. Since the initial value of y is negative, in solving for y the negative square root is used. $\frac{dy}{dx} = \frac{x+1}{y} \Rightarrow y \, dy = (x+1) \, dx$ $\int y \, dy = \int (x+1) \, dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + x + C$ $\frac{(-2)^2}{2} = 0 + 0 + C \Rightarrow 2 = C$ $\frac{y^2}{2} = \frac{x^2}{2} + x + 2 \Rightarrow y^2 = x^2 + 2x + 4$ $y = -\sqrt{x^2 + 2x + 4}$	
(D)	This option is incorrect. All the calculus steps were done correctly as in the calculations for option (C), but the positive square root was used at the last step without consideration of the sign of the initial y value.	

Question 23

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.4D Apply definite integrals to problems involving area, volume, (BC) and length of a curve.	3.4D2: Volumes of solids with known cross sections, including discs and washers, can be calculated with definite integrals.	MPAC 4: Connecting multiple representations MPAC 5: Building notational fluency
(A)	This option is incorrect. This expression uses the entire region between the curves from $x = 0$ to $x = 4$, including the triangle below the x -axis from $x = 0$ to $x = 2$ that is not part of region R . It treats every rotated slice on the entire interval as if forming a washer.	
(B)	This option is incorrect. This expression uses the entire region between the curves from $x = 0$ to $x = 4$, including the triangle below the x -axis from $x = 0$ to $x = 2$ that is not part of region R . It treats every rotated slice on the entire interval as forming a washer, and also makes the mistake of using $\pi(r_2 - r_1)^2$ instead of $\pi(r_2^2 - r_1^2)$ for the area of a washer with inner radius r_1 and outer radius r_2 .	
(C)	This option is correct. The region R must be split into two parts at $x = 2$ when considering its rotation around the x-axis. For the left part, rotating a typical slice around the x-axis will form a disc of radius $r = \sqrt{x}$. For the right part, rotating a typical slice around the x-axis will form a washer with inner radius $r_1 = x - 2$ and outer radius $r_2 = \sqrt{x}$. The combined volume will be given by the integral expression: $\pi \int_0^2 (\sqrt{x})^2 dx + \pi \int_2^4 ((\sqrt{x})^2 - (x - 2)^2) dx$ $= \pi \int_0^2 x dx + \pi \int_2^4 (x - (x - 2)^2) dx.$	
(D)	This option is incorrect. The region was correctly split into two parts at $x = 2$, but the second integral used $\pi(r_2 - r_1)^2$ instead of $\pi(r_2^2 - r_1^2)$ for the area of a washer with inner radius r_1 and outer radius r_2 .	



Question 24

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.1C Determine limits of functions.	1.1C3: Limits of the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$ may be evaluated using L'Hospital's Rule.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. It may have been observed that the numerator is zero at $x = 3$ and the denominator was then ignored.	
(B)	This option is incorrect. The indeterminate form $\frac{0}{0}$ was recognized and an attempt was made to use L'Hospital's Rule, but an error was made in the derivative of the denominator: $\lim_{x \rightarrow 3} \frac{\tan(x-3)}{3e^{x-3} - x} = \lim_{x \rightarrow 3} \frac{\sec^2(x-3)}{3e^{x-3}} = \frac{1}{3}.$	
(C)	This option is correct. Substituting $x = 3$ into the fraction yields the indeterminate form $\frac{0}{0}$. Therefore L'Hospital's Rule can be used to find the limit: $\lim_{x \rightarrow 3} \frac{\tan(x-3)}{3e^{x-3} - x} = \lim_{x \rightarrow 3} \frac{\sec^2(x-3)}{3e^{x-3} - 1} = \frac{1}{2}.$	
(D)	This option is incorrect. The observation that the denominator is zero at $x = 3$ led to the assumption of unbounded behavior, while the numerator was ignored.	

Question 25

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.4A Apply the Mean Value Theorem to describe the behavior of a function over an interval.	2.4A1: If a function f is continuous over the interval $[a, b]$ and differentiable over the interval (a, b) , the Mean Value Theorem guarantees a point within that open interval where the instantaneous rate of change equals the average rate of change over the interval.	MPAC 1: Reasoning with definitions and theorems MPAC 3: Implementing algebraic/computational processes
(A)	This option is incorrect. The error comes from thinking that the Mean Value Theorem guarantees a value of x in the open interval where the instantaneous rate of change of f is equal to the total change in f instead of the average rate of change over the interval. $f'(x) = f(3) - f(1) = 11 - 9 = 2$ $\frac{3x^2 - 6}{x^2} = 2 \Rightarrow 3x^2 - 6 = 2x^2$ $x^2 = 6$ $x = \sqrt{6}$	
(B)	This option is correct. Because f is differentiable for $x > 0$, it is continuous on the closed interval $[1, 3]$ and differentiable on the open interval $(1, 3)$. Therefore the Mean Value Theorem guarantees that there is a number x with $1 < x < 3$ satisfying: $f'(x) = \frac{f(3) - f(1)}{3 - 1} = \frac{11 - 9}{3 - 1} = 1.$ Solve this equation for the value of x in the interval $1 < x < 3$. $\frac{3x^2 - 6}{x^2} = 1 \Rightarrow 3x^2 - 6 = x^2$ $2x^2 = 6$ $x = \sqrt{3}$	
(C)	This option is incorrect. This is the x-value on the interval $1 < x < 3$ where $f'(x) = 0$. The Mean Value Theorem may have been confused with the Extreme Value Theorem.	
(D)	This option is incorrect. This error comes from thinking that the Mean Value Theorem guarantees a value of x in the interval that is equal to the average rate of change over the interval: $x = \frac{f(3) - f(1)}{3 - 1} = \frac{11 - 9}{3 - 1} = 1.$	

Question 26

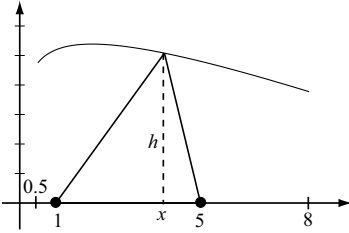
Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.3B(b) Evaluate definite integrals.	3.3B5: Techniques for finding antiderivatives include algebraic manipulation such as long division and completing the square, substitution of variables, (BC) integration by parts, and nonrepeating linear partial fractions. 3.3B2: If f is continuous on the interval $[a, b]$ and F is an antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a).$	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is correct. Before doing the integration, the integrand can be simplified using long division. $\begin{array}{r} x-3 \\ x+2 \overline{) x^2 - x - 5} \\ \underline{x^2 + 2x} \\ -3x - 5 \\ \underline{-3x - 6} \\ 1 \end{array}$ Then: $\int_1^2 \frac{x^2 - x - 5}{x + 2} dx = \int_1^2 \left(x - 3 + \frac{1}{x + 2} \right) dx$ $= \left. \frac{x^2}{2} - 3x + \ln(x + 2) \right _1^2 = (-4 + \ln 4) - \left(-\frac{5}{2} + \ln 3 \right) = -\frac{3}{2} + \ln \frac{4}{3}.$	
(B)	This option is incorrect. The error comes from believing that the definite integral $\int_1^2 \frac{f(x)}{g(x)} dx$ is equal to $\int_1^2 f(x) dx$ divided by $\int_1^2 g(x) dx$. $\frac{\left. \frac{1}{3}x^3 - \frac{1}{2}x^2 - 5x \right _1^2}{\left. \frac{1}{2}x^2 + 2x \right _1^2} = \frac{-\frac{25}{6}}{\frac{7}{2}} = -\frac{25}{21}$	

(C)	This option is incorrect. An algebraic error was made in the long division where terms other than the leading ones were added instead of being subtracted. $ \begin{array}{r} x+1 \\ x+2 \overline{) x^2 - x - 5} \\ \underline{x^2 + 2x} \\ x - 5 \\ \underline{x + 2} \\ -3 \end{array} $ This would lead to: $ \begin{aligned} \int_1^2 \frac{x^2 - x - 5}{x + 2} dx &= \int_1^2 \left(x + 1 - \frac{3}{x + 2} \right) dx \\ &= \left. \frac{x^2}{2} + x - 3 \ln(x + 2) \right _1^2 = (4 - 3 \ln 4) - \left(\frac{3}{2} - 3 \ln 3 \right) = \frac{5}{2} + 3 \ln \frac{3}{4}. \end{aligned} $
(D)	This option is incorrect. The antiderivative of $\frac{f(x)}{g(x)}$ is not the antiderivative of $f(x)$ divided by the antiderivative of $g(x)$. $ \left(\frac{\frac{1}{3}x^3 - \frac{1}{2}x^2 - 5x}{\frac{1}{2}x^2 + 2x} \right) \bigg _1^2 = -\frac{14}{9} + \frac{31}{15} = \frac{23}{45} $

Question 27

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3C Solve problems involving related rates, optimization, rectilinear motion, (BC) and planar motion.	2.3C2: The derivative can be used to solve related rates problems, that is, finding a rate at which one quantity is changing by relating it to other quantities whose rates of change are known.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is correct. The goal in this related rates problem is to find the value of $\frac{dA}{dt}$ at the instant when $h = 3$ and $\frac{dh}{dt} = -\frac{1}{2}$ (the derivative is negative since the depth of the water is decreasing). Using the chain rule gives: $\frac{dA}{dt} = \frac{dA}{dh} \cdot \frac{dh}{dt} = (12\pi - 2\pi h) \frac{dh}{dt}$ $\left. \frac{dA}{dt} \right _{h=3} = (12\pi - 6\pi) \left(-\frac{1}{2} \right) = -3\pi.$ The area is therefore decreasing at the rate of 3π square meters per minute.	
(B)	This option is incorrect. A common mistake in related rates problems is to not use the chain rule, thereby having no $\frac{dh}{dt}$ term to include in the computation: $\frac{dA}{dt} = (12\pi - 2\pi h)$ $\left. \frac{dA}{dt} \right _{h=3} = (12\pi - 6\pi) = 6\pi.$	
(C)	This option is incorrect. Each term in A was differentiated individually, but the chain rule should have been used with the first term because there is no longer an h variable present. The fact that the depth of the water is decreasing also was not taken into account. $\frac{dA}{dt} = 12\pi - 2\pi h \frac{dh}{dt}$ $\left. \frac{dA}{dt} \right _{h=3} = 12\pi - 2\pi(3) \left(\frac{1}{2} \right) = 9\pi$	
(D)	This option is incorrect. This is the value of the area when $h = 3$, not the rate of change of the area.	

Question 28

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3C Solve problems involving related rates, optimization, rectilinear motion, (BC) and planar motion.	2.3C3: The derivative can be used to solve optimization problems, that is, finding a maximum or minimum value of a function over a given interval.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is incorrect. This is the area of the triangle with the third vertex at the endpoint for $x = \frac{1}{2}$: $A = \frac{1}{2}bh = \frac{1}{2}(4)\left(\ln(2x) - \frac{1}{2}x + 5\right) = 2\ln(2x) - x + 10.$ For $x = \frac{1}{2}$, $A = 2\ln\left(2 \cdot \frac{1}{2}\right) - \frac{1}{2} + 10 = 0 - \frac{1}{2} + 10 = \frac{19}{2}.$	
(B)	This option is incorrect. The chain rule was not used in calculating the derivative of the area function. $A = \frac{1}{2}bh = \frac{1}{2}(4)\left(\ln(2x) - \frac{1}{2}x + 5\right) = 2\ln(2x) - x + 10$ $\frac{dA}{dx} = 2\left(\frac{1}{2x}\right) - 1 = \frac{1}{x} - 1$ $\frac{dA}{dx} = \frac{1}{x} - 1 = 0 \Rightarrow x = 1$ $A = 2\ln(2 \cdot 1) - 1 + 10 = 2\ln 2 + 9$	
(C)	This option is correct. The first step is to express the area of the triangle in terms of the variable x . To find the maximum area, the next step is to find the critical value for the area function. $A = \frac{1}{2}bh$ $= \frac{1}{2}(4)\left(\ln(2x) - \frac{1}{2}x + 5\right)$ $= 2\ln(2x) - x + 10$ $\frac{dA}{dx} = 2\left(\frac{2}{2x}\right) - 1 = \frac{2}{x} - 1 = 0 \Rightarrow x = 2 \text{ is the only critical value.}$ For $x = 2$, $A = 2\ln(2 \cdot 2) - 2 + 10 = 2\ln 4 + 8.$ Since there is only one critical value and it is a local maximum by the Second Derivative Test, it must be the absolute maximum. 	

(D)	This option is incorrect. This is the area of the triangle with the third vertex at the endpoint for $x = 8$: $A = \frac{1}{2}bh = \frac{1}{2}(4)\left(\ln(2x) - \frac{1}{2}x + 5\right) = 2\ln(2x) - x + 10.$ For $x = 8$, $A = 2\ln(2 \cdot 8) - 8 + 10 = 2\ln 16 + 2.$
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Question 29

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1C Calculate derivatives.	2.1C6: The chain rule can be used to find the derivative of an inverse function, provided the derivative of that function exists.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency
(A)	This option is incorrect. A misconception that the tangent line for the inverse function is perpendicular to the tangent line for the function could have led to using a slope that is the “negative reciprocal”, i.e., believing that $\frac{d}{dx}f^{-1}(x) = -\frac{1}{f'(x)}$: $f(x) = x^3 + 4x + 2$ $f'(x) = 3x^2 + 4$ $f'(2) = 3 \cdot 2^2 + 4 = 16.$	
(B)	This option is incorrect. The notation $g(x) = f^{-1}(x)$ may have been confused with the reciprocal of $f(x)$, with the attempt then to find $\frac{d}{dx}\frac{1}{f(x)}$ by using the quotient rule (or a power rule along with the chain rule): $\frac{d}{dx}\left(\frac{1}{x^3 + 4x + 2}\right) = \frac{(x^3 + 4x + 2)(0) - (1)(3x^2 + 4)}{(x^3 + 4x + 2)^2}.$ For $x = 2$, this gives: $\frac{-(3 \cdot 2^2 + 4)}{(2^3 + 4 \cdot 2 + 2)^2} = -\frac{16}{18^2} = -\frac{4}{81}.$	

(C)	This option is correct. Since $f(g(x)) = x$, the chain rule can be used to determine that $f'(g(x))g'(x) = 1$. Substituting $x = 2$ gives $1 = f'(g(2))g'(2) = f'(0)g'(2)$. Therefore $g'(2) = \frac{1}{f'(0)} = \frac{1}{3 \cdot 0^2 + 4} = \frac{1}{4}$. Alternatively, one can use the following formula that follows from the same chain rule calculation as above when starting with $f(f^{-1}(x)) = x$: $\left. \frac{d}{dx} f^{-1}(x) \right _{x=2} = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(0)} = \frac{1}{3 \cdot 0^2 + 4} = \frac{1}{4}.$
(D)	This option is incorrect. An attempt might have been made to use the memorized formula for the derivative of an inverse function shown in the solution in option (C), but taking the reciprocal was forgotten, leading to computing $f'(f^{-1}(2))$: $f'(x) = 3x^2 + 4$ $f'(f^{-1}(2)) = f'(0) = 3 \cdot 0^2 + 4 = 4.$

Question 30

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.2A(a) Interpret the definite integral as the limit of a Riemann sum.	3.2A3: The information in a definite integral can be translated into the limit of a related Riemann sum, and the limit of a Riemann sum can be written as a definite integral.	MPAC 1: Reasoning with definitions and theorems MPAC 5: Building notational fluency
(A)	This option is incorrect. This sum used rectangles of width $\Delta x = \frac{1}{n}$ and right endpoints corresponding to step sizes of width $\frac{1}{n}$. These would be the values for a right Riemann sum on the interval $[2, 3]$. This limit is therefore $\int_2^3 x^2 dx$.	
(B)	This option is incorrect. This sum used step sizes of width $\frac{1}{n}$ between the right endpoints which is not consistent with the width $\frac{3}{n}$ used for the rectangles. This limit is $\int_2^3 3x^2 dx$.	
(C)	This option is incorrect. This sum used step sizes of width $\frac{3}{n}$ between the right endpoints which is not consistent with the width $\frac{1}{n}$ used for the rectangles. This limit is $\int_2^5 \frac{1}{3} x^2 dx$.	
(D)	This option is correct. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{5-2}{n} = \frac{3}{n}$. The height of each rectangle is determined by the function x^2 evaluated at the right endpoints of the n intervals. These endpoints are the x -values at $2 + k\Delta x$, for k from 1 to n , since the integral starts at 2. The Riemann sum can be written as $\sum_{k=1}^n (2 + k\Delta x)^2 \Delta x$ where $\Delta x = \frac{3}{n}$, and its limit as $n \rightarrow \infty$ is $\int_2^5 x^2 dx$.	

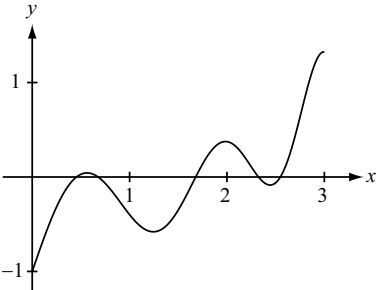
Question 76

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.4E Use the definite integral to solve problems in various contexts.	3.4E1: The definite integral can be used to express information about accumulation and net change in many applied contexts.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
(A)	This option is correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $s(t)$ is the rate of change at which sand is added to the beach, the definite integral of $s(t)$ on the interval $2 \leq t \leq 5$ gives the net change in the amount of sand on the beach over that time period. Since t is measured in hours since 5:00 A.M., the time 7:00 A.M. corresponds to $t = 2$ and 10:00 A.M. corresponds to $t = 5$. $\int_2^5 s(t) \, dt = \int_2^5 (65 + 24 \sin(0.3t)) \, dt = 255.368$	
(B)	This option is incorrect. The definite integral was correctly used to find a net change given a rate of change, but incorrectly uses the 3-hour period from $t = 0$ to $t = 3$ by not adjusting for the variable t representing the time since 5:00 A.M. $\int_0^3 s(t) \, dt$	
(C)	This option is incorrect. This is the average value of the rate at which sand is being added over the interval $2 \leq t \leq 5$: $\frac{\int_2^5 s(t) \, dt}{5 - 2}.$	
(D)	This option is incorrect. This is the net change $s(5) - s(2)$ in the rate at which sand is being added over the interval $2 \leq t \leq 5$, not the net change in the amount of sand on the beach over that interval.	

Question 77

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.1B Estimate limits of functions.	1.1B1: Numerical and graphical information can be used to estimate limits.	MPAC 1: Reasoning with definitions and theorems MPAC 4: Connecting multiple representations
(A)	This option is incorrect. The value $a = 4$ is missed, perhaps because of a mistaken belief that the function must be continuous at the point for the limit to exist.	
(B)	This option is correct. $\lim_{x \rightarrow a} f(x) = 0$ if the left-sided and right-sided limits at $x = a$ both equal 0, i.e. $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = 0$. For this graph, that happens at $x = 2$ and $x = 4$. Note that the function does not need to be defined at $x = 4$ for the limit to exist there.	
(C)	This option is incorrect. These are the values for which $f(x) = 0$ and at least a one-sided limit exists. However, $\lim_{x \rightarrow a} f(x)$ does not exist for $a = 0$.	
(D)	This option is incorrect. These are the values where $f(x) = 0$. However, $\lim_{x \rightarrow a} f(x)$ does not exist for $a = 0$ and the limit is equal to -1 , not 0 , for $a = 1$.	

Question 78

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.2A Use derivatives to analyze properties of a function.	2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is incorrect. This might be selected if a badly chosen viewing window (i.e. not examining the correct interval) omits the first two and last two intersections.	
(B)	This option is incorrect. This might be selected if the viewing window omits the first two intersections or the last two intersections, or if the first two intersections are overlooked, perhaps because the range on the y -axis is set too large.	
(C)	This option is incorrect. This is the apparent number of points of inflection on the graph of f'' . It is also the number of critical points of f'' on the open interval $0 < x < 3$.	
(D)	This option is correct. Graph $f''(x)$ on the interval $0 \leq x \leq 3$.  A point of inflection occurs where the second derivative $f''(x)$ changes sign. This happens five times on the interval: twice between 0 and 1, once between 1 and 2, and twice between 2 and 3.	

Question 79

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.4C Apply definite integrals to problems involving motion.	3.4C1: For a particle in rectilinear motion over an interval of time, the definite integral of velocity represents the particle's displacement over the interval of time, and the definite integral of speed represents the particle's total distance traveled over the interval of time.	MPAC 4: Connecting multiple representations MPAC 1: Reasoning with definitions and theorems
(A)	This option is incorrect. This is the difference between the displacement and the total distance over the whole interval $0 \leq t \leq 5$.	
(B)	This option is correct. Let the total area above the t -axis be A and the area below the t -axis be B . The particle's total distance on the time interval $0 \leq t \leq 5$ is: $\int_0^5 v(t) dt = A + B = 13,$ and the particle's displacement is: $\int_0^5 v(t) dt = A - B = 3.$ Solving the system gives $A = 8$ and $B = 5$. Since the graph of $v(t)$ is below the axis for $2 < t < 4$, $\int_2^4 v(t) dt = -B = -5.$	
(C)	This option is incorrect. $\int_2^4 v(t) dt = B = 5$ was concluded by thinking just about area and not that the definite integral corresponds to a region below the axis.	
(D)	This option is incorrect. This is the difference between the total distance and the displacement over the whole interval $0 \leq t \leq 5$.	

Question 80

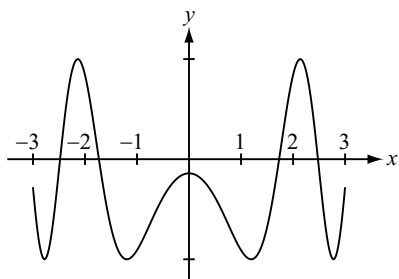
Learning Objective		Essential Knowledge	Mathematical Practices for AP Calculus
3.4A Interpret the meaning of a definite integral within a problem.		3.4A2: The definite integral of the rate of change of a quantity over an interval gives the net change of that quantity over that interval.	MPAC 2: Connecting concepts MPAC 5: Building notational fluency
(A)	This option is incorrect. If $T(t)$ was the room temperature, this would be the interpretation for the expression $T(0) + \int_0^6 H(t) dt$ that would be used to find the temperature at $t = 6$ given the initial value of 20.		
(B)	This option is incorrect. If $T(t)$ was the room temperature, this would be an interpretation for the expression $\frac{1}{6} \int_0^6 T(t) dt$.		
(C)	This option is correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $H(t)$ is the rate of change of temperature, the definite integral of $H(t)$ on the interval $0 \leq t \leq 6$ gives the change in the temperature over that time period.		
(D)	This option is incorrect. This is an interpretation of $H(6)$.		

Question 81

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.2A Use derivatives to analyze properties of a function.	2.2A3: Key features of the graphs of f , f' , and f'' are related to one another.	MPAC 4: Connecting multiple representations MPAC 2: Connecting concepts
(A)	This option is correct. Where the derivative f' is positive, the function f should be increasing. Where the derivative f' is negative, the function f should be decreasing. The derivative to the left of $x = 0$ is always positive. The derivative to the right of $x = 0$ is positive until $x = 1$, then negative. Therefore the graph of f should be increasing for $x < 0$ and $0 < x < 1$, and should be decreasing for $x > 1$. The graph in this option could therefore be the graph of f .	
(B)	This option is incorrect. The sign of the derivative on the left side and right side of $x = 0$ was misinterpreted, thus reversing the desired increasing/decreasing behavior that the graph of f should display.	
(C)	This option is incorrect. The sign of the derivative on the left side of $x = 0$ was misinterpreted, thus reversing the desired increasing behavior that the graph of f should display on the left side of $x = 0$.	
(D)	This option is incorrect. The sign of the derivative on the right side of $x = 0$ was misinterpreted, thus not achieving the desired increasing/decreasing behavior that the graph of f should display on the right side of $x = 0$.	

Question 82

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.2A Use derivatives to analyze properties of a function.	2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is correct. The graph of f' on the interval $-3 < x < 3$ shows that the function f will have a relative maximum where the derivative changes from positive to negative. This happens between -2 and -1, and between 2 and 3. The calculator is used to find the zeros of f' on those intervals.	
(B)	This option is incorrect. These are the values where the derivative changes from negative to positive. They would be where f has a relative minimum, not a relative maximum.	
(C)	This option is incorrect. These are the values where the derivative has a relative maximum, not where the function has a relative maximum.	
(D)	This option is incorrect. These are all of the values where the derivative is equal to zero. They are critical values, but the change in the sign of the derivative has not been used to distinguish between a relative maximum and a relative minimum.	



Question 83

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.2A Analyze functions for intervals of continuity or points of discontinuity.	1.2A3: Types of discontinuities include removable discontinuities, jump discontinuities, and discontinuities due to vertical asymptotes.	MPAC 4: Connecting multiple representations MPAC 1: Reasoning with definitions and theorems
(A)	This option is incorrect. This might be chosen because the function f is not differentiable at $x = 1$. However, it is continuous at this point.	
(B)	This option is incorrect. The type of discontinuity was misidentified. Since $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 0$, there is no jump discontinuity at $x = 3$. The function has a removable discontinuity at $x = 3$.	
(C)	This option is correct. A jump discontinuity occurs at $x = c$ if the left- and right-sided limits exist at c , but are not equal, that is, $\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x).$ For this function, $\lim_{x \rightarrow 7^-} f(x) = 4$ $\lim_{x \rightarrow 7^+} f(x) = 1$ and so there is a jump discontinuity at $x = 7$.	
(D)	This option is incorrect. The type of discontinuity was misidentified. Since $\lim_{x \rightarrow 10^-} f(x) = \lim_{x \rightarrow 10^+} f(x) \approx 0.5$, there is no jump discontinuity at $x = 10$. The function is defined at $x = 10$ and has a removable discontinuity there.	

Question 84

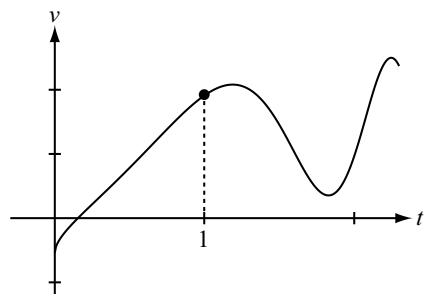
Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
3.3B(b) Evaluate definite integrals.	3.3B2: If f is continuous on the interval $[a, b]$ and F is an antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
(A)	This option is incorrect. A misunderstanding of the Fundamental Theorem of Calculus resulted in computing $f(1) + \int_1^5 f''(x) dx$.	
(B)	This option is incorrect. Computing $f(1) + f'(5)$ shows a misunderstanding of the Fundamental Theorem of Calculus.	
(C)	This option is incorrect. A common error is to neglect to use the initial condition and just calculate $f(5) = \int_1^5 \sqrt{x^3 + 6} dx$. This may arise from a misunderstanding of the Fundamental Theorem of Calculus or just forgetting to add the initial condition after computing the definite integral with the calculator.	
(D)	This option is correct. By the Fundamental Theorem of Calculus, $f(5) - f(1) = \int_1^5 f'(x) dx$. Therefore $f(5) = f(1) + \int_1^5 \sqrt{x^3 + 6} dx = \pi + 24.672 \approx 27.814$.	

Question 85

Learning Objective		Essential Knowledge	Mathematical Practices for AP Calculus
2.3D Solve problems involving rates of change in applied contexts.		2.3D1: The derivative can be used to express information about rates of change in applied contexts.	MPAC 1: Reasoning with definitions and theorems MPAC 2: Connecting concepts
(A)	This option is incorrect. The focus was only on the rate at which people enter the building. This inequality says that the total number of people who have entered the building is increasing.		
(B)	This option is incorrect. Here the focus is only on the rate at which people enter the building. This inequality says that the rate of change of the number of people entering the building is increasing.		
(C)	This option is incorrect. The expression $R(t) = f(t) - g(t)$ indicates the rate of change of the number of people in the building (rate in minus rate out). The sign of $R(t)$ determines whether the number of people in the building is increasing or decreasing at time t . This inequality says that the number of people in the building is increasing.		
(D)	This option is correct. The expression $R(t) = f(t) - g(t)$ indicates the rate of change of the number of people in the building (rate in minus rate out). If the derivative $R'(t) = f'(t) - g'(t)$ is positive, then $R(t)$ is increasing, i.e. the rate of change of the number of people in the building is increasing.		

Question 86

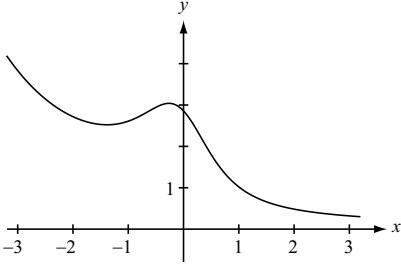
Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3C Solve problems involving related rates, optimization, rectilinear motion, (BC) and planar motion.	2.3C1: The derivative can be used to solve rectilinear motion problems involving position, speed, velocity, and acceleration.	MPAC 1: Reasoning with definitions and theorems MPAC 2: Connecting concepts
(A)	This option is incorrect. The meaning of the sign of the velocity may have been misinterpreted.	
(B)	This option is correct. At $t = 1$, the graph of v is positive. In addition, the graph of v is increasing at $t = 1$, which means that $v'(1)$ is positive. Alternatively, $v(1) = 1.912$ and $v'(1) = 1.617$. Since velocity is the derivative of position, a positive value of v at $t = 1$ means that x is increasing, i.e. the particle is moving to the right. Since acceleration is the derivative of velocity, a positive value of v' at $t = 1$ means that the particle is moving with positive acceleration.	
(C)	This option is incorrect. Both claims are reversed.	
(D)	This option is incorrect. The meaning of the sign of the derivative of velocity may have been misinterpreted.	



Question 87

Learning Objective		Essential Knowledge	Mathematical Practices for AP Calculus
2.3A Interpret the meaning of a derivative within a problem.		2.3A1: The unit for $f'(x)$ is the unit for f divided by the unit for x .	MPAC 5: Building notational fluency MPAC 3: Implementing algebraic/computational processes
(A)	This option is incorrect. This is the units of the dependent variable only.		
(B)	This option is incorrect. This is the units of the independent variable only.		
(C)	This option is incorrect. The dependent and independent variables were inverted. While in many problems time is often the independent variable, in this model time is specified as a function of the air pressure and hence is the dependent variable. That is why it is important to understand the functional notation $t = f(p)$.		
(D)	This option is correct. The derivative of the function $t = f(p)$ is the limit of the difference quotient $\frac{\Delta t}{\Delta p}$. The units on the dependent variable, t , in the numerator is hours and the units on the independent variable, p , in the denominator is psi. The rate of change, therefore, has units in hours per psi.		

Question 88

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.2A Use derivatives to analyze properties of a function.	2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection.	MPAC 3: Implementing algebraic/computational processes MPAC 2: Connecting concepts
(A)	This option is incorrect. A common error is to answer the question as if the graph drawn on the calculator were the graph of the function f , not the graph of the derivative f' . The function f' is increasing between its local minimum and its local maximum, and this interval appears to match the location of those two points. The precise values can be found by solving for where $f''(x) = 0$.	
(B)	This option is incorrect. There may have been confusion about which property of the derivative provides information about where the function is increasing. If one believes that the desired property is where the graph of f' is concave up, then these two intervals appear to work. The precise values of the endpoints can be found by solving for where $f'''(x) = 0$.	
(C)	This option is correct. The function f will be increasing where f' is positive. Graph f' on an interval that includes at least $-2 \leq x \leq 1$ since that covers the choices in options (A) and (B).  The graph of f' is always positive on the interval in the window. The graph therefore rules out options (A), (B), and (D). The best choice is therefore (C). [One cannot verify graphically that f' is positive for $-\infty < x < \infty$. The denominator in the formula for f' is always positive. The numerator is clearly positive for $x \geq 0$. For $x < 0$, observe that the graph of $2e^{-x}$ is concave up and therefore lies above the tangent line at $x = 0$. This tangent line is $y = 2 - 2x$. Hence $2e^{-x} > 2 - 2x > -x$ since $x < 0 < 2$.]	
(D)	This option is incorrect. There may have been confusion about which sign of the derivative provides information about the increasing behavior of the function. Or the observation that there are no zeros of f' might incorrectly suggest that there are no intervals to investigate.	

Question 89

Learning Objective		Essential Knowledge	Mathematical Practices for AP Calculus
1.2B Determine the applicability of important calculus theorems using continuity.		1.2B1: Continuity is an essential condition for theorems such as the Intermediate Value Theorem, the Extreme Value Theorem, and the Mean Value Theorem.	MPAC 1: Reasoning with definitions and theorems MPAC 4: Connecting multiple representations
(A)	This option is incorrect. The number of times that $f(x) = 4$ may have been confused with the number of times that $x = 4$, or it may have been thought that the Intermediate Value Theorem only applies to the interval $[a, b]$ when $f(a) \geq 4 \geq f(b)$, namely on $[4, 6]$.		
(B)	This option is incorrect. It may have been thought that the Intermediate Value Theorem only applies to the interval $[a, b]$ when $f(a) \leq 4 \leq f(b)$, namely on $[0, 4]$ and $[8, 13]$.		
(C)	This option is correct. Because f is continuous, the Intermediate Value Theorem will imply that $f(x)$ will equal 4 on an interval $[a, b]$ where 4 is between $f(a)$ and $f(b)$. According to the table, this happens for the intervals $[0, 4]$, $[4, 6]$, and $[8, 13]$. So there are at least three values of x for which the Intermediate Value Theorem applies and $f(x) = 4$.		
(D)	This option is incorrect. It may have been thought that an additional value of x occurs on $[6, 13]$ because $f(6) < 4 < f(13)$. But this x -value may be the same as the one guaranteed by the Intermediate Value Theorem on the interval $[8, 13]$.		

Question 90

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.2B Recognize the connection between differentiability and continuity.	2.2B2: If a function is differentiable at a point, then it is continuous at that point.	MPAC 1: Reasoning with definitions and theorems MPAC 4: Connecting multiple representations
(A)	This option is incorrect. It does not recognize that statement II is true.	
(B)	This option is correct. By the Fundamental Theorem of Calculus, $h(3) = h(-1) + \int_{-1}^3 h'(x) dx$. Since the region between the graph of h' and the x-axis has more area above the axis than below the axis, the definite integral $\int_{-1}^3 h'(x) dx$ is positive. Therefore $h(3) > h(-1)$ and statement I is false. The graph shows that derivative of h exists on the interval $(-1, 3)$. That means that the function h is differentiable on $(-1, 3)$ and is therefore continuous on that interval. In particular, h is continuous at $x = 1$, so statement II must be true. Because h is continuous at $x = 1$ and $h'(1) \approx 0.5$, the graph of h has a tangent line at $x = 1$ that is not vertical. Locally near $x = 1$ the graph of h will look like that tangent line and thus there cannot be a vertical asymptote at $x = 1$. Therefore statement III is false.	
(C)	This option is incorrect. Statement I might be chosen by interpreting the given graph as the graph of h rather than its derivative. It might also be chosen by recognizing that $h(3) - h(-1) = \int_{-1}^3 h'(x) dx$ but believing that the symmetry of the areas within the semicircles results in a net change of 0 for h over the interval $-1 \leq x \leq 3$.	
(D)	This option is incorrect. Statement I might be chosen for the same reasons as in option (C). Statement III might be chosen by interpreting the given graph as the graph of h rather than its derivative and confusing vertical asymptote with vertical tangent, or by thinking that the vertical tangent on h' at $x = 1$ results in a vertical asymptote on h at $x = 1$.	

Section II: Free Response

Free-response questions provide students with an opportunity to demonstrate their knowledge of correct mathematical reasoning and thinking. In most cases, an answer without supporting work will receive no credit; students are required to articulate the reasoning and methods that support their answer. Some questions will ask students to justify an answer or discuss whether a theorem can be applied. There are two free-response questions in Part A and four questions in Part B. Each part of the free-response section is timed, and students may use a graphing calculator only for Part A. During the timed portion for Part B of the free-response section, students are allowed to return to working on Part A questions, though without the use of a graphing calculator.

Curriculum Framework Alignment for Free-Response Question 1

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.1B Estimate derivatives.	2.1B1: The derivative at a point can be estimated from information given in tables or graphs.	MPAC 1: Reasoning with definitions and theorems
2.3A Interpret the meaning of a derivative within a problem.	2.3A1: The unit for $f'(x)$ is the unit for f divided by the unit for x . 2.3A2: The derivative of a function can be interpreted as the instantaneous rate of change with respect to its independent variable.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
2.3D Solve problems involving rates of change in applied contexts.	2.3D1: The derivative can be used to express information about rates of change in applied contexts.	MPAC 4: Connecting multiple representations
3.2B Approximate a definite integral.	3.2B2: Definite integrals can be approximated using a left Riemann sum, a right Riemann sum, a midpoint Riemann sum, or a trapezoidal sum; approximations can be computed using either uniform or nonuniform partitions.	MPAC 5: Building notational fluency MPAC 6: Communicating
3.4B Apply definite integrals to problems involving the average value of a function.	3.4B1: The average value of a function f over an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$.	

Scoring Guidelines for Free-Response Question 1

Question 1

(a) $h'(6) \approx \frac{h(7) - h(5)}{7 - 5} = \frac{18.5 - 15.5}{2} = 1.5 \text{ in/min}$

1 : answer with units

(b) $\int_0^{10} h(t) dt \approx (2 - 0) \cdot h(2) + (5 - 2) \cdot h(5) + (7 - 5) \cdot h(7) + (10 - 7) \cdot h(10)$
 $= 2(10.0) + 3(15.5) + 2(18.5) + 3(20.0) = 163.5$

3 : $\begin{cases} 1 : \text{right Riemann sum} \\ 1 : \text{approximation} \\ 1 : \text{overestimate} \\ \text{with reason} \end{cases}$

Because h is an increasing function, the right Riemann sum approximation is greater than $\int_0^{10} h(t) dt$.

(c) Average depth in tank $B = \frac{1}{10} \int_0^{10} g(t) dt = 16.624 \text{ in}$

Average depth in tank $A = \frac{1}{10} \int_0^{10} h(t) dt < \frac{1}{10}(163.5) = 16.35 \text{ in} < 16.624 \text{ in}$

3 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{average depth} \\ \text{in tank } B \\ 1 : \text{answer with reason} \end{cases}$

Therefore, the average depth of the water in tank B is greater than the average depth of the water in tank A .

(d) $g'(6) = 0.887$

2 : $\begin{cases} 1 : \text{uses } g'(6) \\ 1 : \text{answer with reason} \end{cases}$

The depth of the water in tank B is increasing at time $t = 6$ because $g'(6) > 0$.

Curriculum Framework Alignment for Free-Response Question 2

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.2B Determine the applicability of important calculus theorems using continuity.	1.2B1: Continuity is an essential condition for theorems such as the Intermediate Value Theorem, the Extreme Value Theorem, and the Mean Value Theorem.	MPAC 1: Reasoning with definitions and theorems MPAC 2: Connecting concepts
2.2A Use derivatives to analyze properties of a function.	2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection.	MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency MPAC 6: Communicating
2.3C Solve problems involving related rates, optimization, rectilinear motion, (BC) and planar motion.	2.3C1: The derivative can be used to solve rectilinear motion problems involving position, speed, velocity, and acceleration.	
2.4A Apply the Mean Value Theorem to describe the behavior of a function over an interval.	2.4A1: If a function f is continuous over the interval $[a, b]$ and differentiable over the interval (a, b) , the Mean Value Theorem guarantees a point within that open interval where the instantaneous rate of change equals the average rate of change over the interval.	
3.4C Apply definite integrals to problems involving motion.	3.4C1: For a particle in rectilinear motion over an interval of time, the definite integral of velocity represents the particle's displacement over the interval of time, and the definite integral of speed represents the particle's total distance traveled over the interval of time.	

Question 2

- (a)
- $a_Q(t) > 0$
- for
- $0 < t < 3.963$
- .

The velocity of particle Q is increasing on the interval $(0, 3.963]$.

$$2 : \begin{cases} 1 : \text{interval} \\ 1 : \text{reason} \end{cases}$$

$$\begin{aligned} \text{(b)} \quad x_Q(3) &= x_Q(0) + \int_0^3 v_Q(t) \, dt \\ &= 2 + (-3.490102) \\ &= -1.490 \end{aligned}$$

$$3 : \begin{cases} 1 : \text{integral} \\ 1 : \text{uses } x_Q(0) \\ 1 : \text{answer} \end{cases}$$

- (c)
- $x_R(t)$
- is differentiable.
- $\Rightarrow x_R(t)$
- is continuous on
- $1 \leq t \leq 3$
- .

$$\frac{x_R(3) - x_R(1)}{3 - 1} = \frac{8 - 4}{2} = 2$$

$$2 : \begin{cases} 1 : \frac{x_R(3) - x_R(1)}{3 - 1} \\ 1 : \text{conclusion, using MVT} \end{cases}$$

Therefore, by the Mean Value Theorem, there is at least one time t , $1 < t < 3$, for which $x'_R(t) = v_R(t) = 2$.

- (d)
- $x_Q(3) = -1.490$
- and
- $v_Q(3) = 1.682$
- (or 1.681)

Particle Q is moving to the right from $x = -1.490$.

$$x_R(3) = 8 \text{ and } v_R(3) = -2$$

Particle R is moving to the left from $x = 8$.

Therefore, the particles are moving toward each other.

$$2 : \begin{cases} 1 : \text{answer} \\ 1 : \text{explanation} \end{cases}$$

Curriculum Framework Alignment for Free-Response Question 3

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.1C Determine limits of functions.	1.1C3: Limits of the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$ may be evaluated using L'Hospital's Rule.	MPAC 1: Reasoning with definitions and theorems MPAC 2: Connecting concepts
2.1C Calculate derivatives.	2.1C3: Sums, differences, products, and quotients of functions can be differentiated using derivative rules.	MPAC 3: Implementing algebraic/computational processes
2.2A Use derivatives to analyze properties of a function.	2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection. 2.2A3: Key features of the graphs of f , f' , and f'' are related to one another.	MPAC 4: Connecting multiple representations MPAC 5: Building notational fluency MPAC 6: Communicating
3.2C Calculate a definite integral using areas and properties of definite integrals.	3.2C1: In some cases, a definite integral can be evaluated by using geometry and the connection between the definite integral and area.	
3.3A Analyze functions defined by an integral.	3.3A2: If f is a continuous function on the interval $[a, b]$, then $\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x),$ where x is between a and b . 3.3A3: Graphical, numerical, analytical, and verbal representations of a function f provide information about the function g defined as $g(x) = \int_a^x f(t) dt.$	

Question 3

(a) $f'(x) = g(x)$

The function f has critical points at $x = -6$ and $x = 2$.

f has a local minimum at $x = -6$ because f' changes from negative to positive at that value.

f has a local maximum at $x = 2$ because f' changes from positive to negative at that value.

(b) $f(0) = \int_{-8}^0 g(t) dt = 8 + 4\pi$

(c) $\lim_{x \rightarrow -4} f(x) = f(-4) = \int_{-8}^{-4} g(t) dt = 0$
 $\lim_{x \rightarrow -4} (x^2 + 4x) = 0$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -4} \frac{f(x)}{x^2 + 4x} = \lim_{x \rightarrow -4} \frac{f'(x)}{2x + 4} = \lim_{x \rightarrow -4} \frac{g(x)}{2x + 4} = \frac{2}{-4} = -\frac{1}{2}$$

(d) $h'(x) = \frac{(x^2 + 1)g'(x) - g(x)2x}{(x^2 + 1)^2}$

$$\begin{aligned} h'(1) &= \frac{(1^2 + 1)g'(1) - g(1)(2)(1)}{(1^2 + 1)^2} \\ &= \frac{2(-3) - 3(2)}{4} = -3 \end{aligned}$$

$$3 : \begin{cases} 1 : f'(x) = g(x) \\ 1 : \text{critical points} \\ 1 : \text{classifications with justification} \end{cases}$$

1 : answer

$$2 : \begin{cases} 1 : \text{L'Hospital's Rule} \\ 1 : \text{answer} \end{cases}$$

$$3 : \begin{cases} 2 : h'(x) \\ 1 : \text{answer} \end{cases}$$

Curriculum Framework Alignment for Free-Response Question 4

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.1C Determine limits of functions.	1.1C1: Limits of sums, differences, products, quotients, and composite functions can be found using the basic theorems of limits and algebraic rules.	MPAC 1: Reasoning with definitions and theorems MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes MPAC 5: Building notational fluency MPAC 6: Communicating
2.1C Calculate derivatives.	2.1C3: Sums, differences, products, and quotients of functions can be differentiated using derivative rules. 2.1C5: The chain rule is the basis for implicit differentiation.	
2.1D Determine higher order derivatives.	2.1D2: Higher order derivatives are represented with a variety of notations. For $y = f(x)$, notations for the second derivative include $\frac{d^2y}{dx^2}$, $f''(x)$, and y'' . Higher order derivatives can be denoted $\frac{d^n y}{dx^n}$ or $f^{(n)}(x)$.	
2.2A Use derivatives to analyze properties of a function.	2.2A1: First and second derivatives of a function can provide information about the function and its graph including intervals of increase or decrease, local (relative) and global (absolute) extrema, intervals of upward or downward concavity, and points of inflection.	
3.5A Analyze differential equations to obtain general and specific solutions.	3.5A2: Some differential equations can be solved by separation of variables.	

Question 4

(a) $\frac{dy}{dx} = (y - 2)(x^2 + 1)$

$$\int \frac{dy}{y - 2} = \int (x^2 + 1) dx$$

$$\ln|y - 2| = \frac{x^3}{3} + x + C$$

$$\ln 3 = \frac{0^3}{3} + 0 + C \Rightarrow C = \ln 3$$

Because $y(0) = 5$, $y > 2$, so $|y - 2| = y - 2$.

$$y - 2 = 3e^{\frac{x^3}{3} + x}$$

$$y = 2 + 3e^{\frac{x^3}{3} + x}$$

Note: this solution is valid for all real numbers.

(b) $\lim_{x \rightarrow -\infty} \left(2 + 3e^{\frac{x^3}{3} + x} \right) = 2$

(c) $\left. \frac{dy}{dx} \right|_{(1,3)} = (3 - 2)(1^2 + 1) = 2$

$$\frac{d^2y}{dx^2} = \frac{dy}{dx}(x^2 + 1) + (y - 2)(2x)$$

$$\left. \frac{d^2y}{dx^2} \right|_{(1,3)} = (2)(1^2 + 1) + (3 - 2)(2) = 6$$

Because $\left. \frac{d^2y}{dx^2} \right|_{(1,3)} > 0$ and $\frac{d^2y}{dx^2}$ is continuous, the graph of

$y = f(x)$ is concave up at the point $(1, 3)$.

$$5 : \begin{cases} 1 : \text{separation of variables} \\ 2 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ \text{and uses initial condition} \\ 1 : \text{solves for } y \end{cases}$$

Note: max 3/5 [1-2-0-0] if no constant of integration

Note: 0/5 if no separation of variables

1 : answer

$$3 : \begin{cases} 2 : \left. \frac{d^2y}{dx^2} \right|_{(1,3)} \\ 1 : \text{concave up with reason} \end{cases}$$

Curriculum Framework Alignment for Free-Response Question 5

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
1.1A(a) Express limits symbolically using correct notation.	1.1A2: The concept of a limit can be extended to include one-sided limits, limits at infinity, and infinite limits.	MPAC 1: Reasoning with definitions and theorems
1.2A Analyze functions for intervals of continuity or points of discontinuity.	1.2A1: A function f is continuous at $x = c$ provided that $f(c)$ exists, $\lim_{x \rightarrow c} f(x)$ exists, and $\lim_{x \rightarrow c} f(x) = f(c)$.	MPAC 2: Connecting concepts MPAC 3: Implementing algebraic/computational processes
2.1C Calculate derivatives.	2.1C2: Specific rules can be used to calculate derivatives for classes of functions, including polynomial, rational, power, exponential, logarithmic, trigonometric, and inverse trigonometric.	MPAC 5: Building notational fluency MPAC 6: Communicating
2.2B Recognize the connection between differentiability and continuity.	2.2B2: If a function is differentiable at a point, then it is continuous at that point.	
3.2C Calculate a definite integral using areas and properties of definite integrals.	3.2C2: Properties of definite integrals include the integral of a constant times a function, the integral of the sum of two functions, reversal of limits of integration, and the integral of a function over adjacent intervals. 3.2C3: The definition of the definite integral may be extended to functions with removable or jump discontinuities.	
3.3B(b) Evaluate definite integrals.	3.3B2: If f is continuous on the interval $[a, b]$ and F is an antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$.	

Question 5

$$(a) \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (3x^2 + 2x) = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (e^{2x} + 2) = 3$$

Therefore, $\lim_{x \rightarrow 0} f(x)$ does not exist, and f is not continuous at $x = 0$.

$$(b) \quad f'(-2) = (6x + 2)|_{x=-2} = 6(-2) + 2 = -10$$

$$f'(3) = 2e^{2x}|_{x=3} = 2e^{(2)(3)} = 2e^6$$

(c) $f'(0)$ does not exist because f is not continuous at $x = 0$.

$$\begin{aligned} (d) \quad g(1) &= \int_{-1}^1 f(t) \, dt = \int_{-1}^0 (3t^2 + 2t) \, dt + \int_0^1 (e^{2t} + 2) \, dt \\ &= \left[t^3 + t^2 \right]_{-1}^0 + \left[\frac{e^{2t}}{2} + 2t \right]_0^1 \\ &= [(0 + 0) - (-1 + 1)] + \left[\left(\frac{e^2}{2} + 2 \right) - \left(\frac{1}{2} + 0 \right) \right] \\ &= \frac{e^2}{2} + \frac{3}{2} \end{aligned}$$

2 : $\begin{cases} 1 : \text{considers one-sided limits} \\ 1 : \text{answer with justification} \end{cases}$

2 : $\begin{cases} 1 : f'(-2) \\ 1 : f'(3) \end{cases}$

1 : explanation

4 : $\begin{cases} 1 : \text{integrals} \\ 2 : \text{antiderivatives} \\ 1 : \text{answer} \end{cases}$

Curriculum Framework Alignment for Free-Response Question 6

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
2.3C Solve problems involving related rates, optimization, rectilinear motion, (BC) and planar motion.	2.3C3: The derivative can be used to solve optimization problems, that is, finding a maximum or minimum value of a function over a given.	MPAC 1: Reasoning with definitions and theorems MPAC 2: Connecting concepts
3.1A Recognize antiderivatives of basic functions.	3.1A2: Differentiation rules provide the foundation for finding antiderivatives.	MPAC 3: Implementing algebraic/computational processes
3.3A Analyze functions defined by an integral.	3.3A1: The definite integral can be used to define new functions; for example, $f(x) = \int_0^x e^{-t^2} dt.$ 3.3A2: If f is a continuous function on the interval $[a, b]$, then $\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$, where x is between a and b .	MPAC 5: Building notational fluency MPAC 6: Communicating
3.3B(b) Evaluate definite integrals.	3.3B2: If f is continuous on the interval $[a, b]$ and F is an antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a).$	
3.4A Interpret the meaning of a definite integral within a problem.	3.4A1: A function defined as an integral represents an accumulation of a rate of change.	
3.4E Use the definite integral to solve problems in various contexts.	3.4E1: The definite integral can be used to express information about accumulation and net change in many applied contexts.	

Question 6

$$(a) \int_0^2 L(t) dt = \int_0^2 (-2t + 15) dt = [-t^2 + 15t]_0^2 \\ = -4 + 30 = 26$$

26 hundred bees leave the hive during the time interval $0 \leq t \leq 2$.

(b) The total number of bees, in hundreds, in the hive at time t is

$$35 + \int_0^t (E(x) - L(x)) dx.$$

$$\begin{aligned} 35 + \int_0^4 (E(x) - L(x)) dx &= 35 + \int_0^4 ((16x - 3x^2) - (-2x + 15)) dx \\ &= 35 + \int_0^4 (-3x^2 + 18x - 15) dx \\ &= 35 + [-x^3 + 9x^2 - 15x]_0^4 \\ &= 35 + (-64 + 144 - 60) \\ &= 55 \end{aligned}$$

55 hundred bees are in the hive at time $t = 4$.

(c) Let $B(t)$ be the total number of bees, in hundreds, in the hive at time t , for $0 \leq t \leq 4$.

$$B(t) = 35 + \int_0^t (E(x) - L(x)) dx$$

$$B'(t) = E(t) - L(t) = -3t^2 + 18t - 15 = -3(t-1)(t-5)$$

$$B'(t) = 0 \Rightarrow t = 1, t = 5$$

t	$B(t)$
0	35
1	28
4	55

The minimum numbers of bees in the hive for $0 \leq t \leq 4$ is 28 hundred bees.

3 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$

3 : $\begin{cases} 1 : \text{expression for total} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$

3 : $\begin{cases} 1 : \text{sets } E(t) - L(t) = 0 \\ 1 : \text{answer} \\ 1 : \text{justification} \end{cases}$